Information in Disclosing Emerging Technologies: Evidence from AI Disclosure

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Abstract

We examine the information content of voluntary artificial intelligence (AI) disclosures in corporate annual reports. Utilizing a novel approach that integrates traditional keyword-based textual analysis with large language models, we classify AI-related disclosures into categories such as revenue generation, cost reduction, and various AI-related risks. Our analysis reveals that AI disclosure has significantly increased across all industry sectors since 2010 and strongly correlates with a firm's actual AI activities, as indicated by AI-related job postings. We also find that AI revenue and cost disclosures provide additional predictive information beyond a firm's current AI activities, offering insights into future growth, investment, and operational efficiency. Furthermore, AI risk disclosures are associated with higher tail risk for firms. Our findings suggest that voluntary AI disclosures reflect a firm's current engagement in AI and offer valuable forward-looking information about the risks and opportunities linked to AI, useful for external stakeholders in assessing a firm's AI involvement.

Keywords: Artificial Intelligence, 10-K disclosure, Firm Growth, Investment, Risk

JEL Classification: D22, E22, G14, G32, M41

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1. Introduction

Artificial Intelligence (AI) has emerged as one of the most transformative technologies shaping today's business landscapes. A recent survey of 240 chief executive officers and senior executives shows that almost 60% of business leaders anticipate AI will "fundamentally transform the landscape."[1] As a result, this emerging, widespread, and rapidly evolving technology introduces significant uncertainty for investors when assessing its potential impact on firm value. On one hand, AI may enhance future cash flows by driving revenue growth through the creation of innovative products and more precise customer targeting. Additionally, AI can reduce operational costs by automating routine tasks, improving supply chain management, and optimizing production processes, thereby boosting overall efficiency. On the other hand, AI adoption also introduces heightened risks. One major source of risk is the evolving regulatory landscape surrounding AI, as policymakers grapple with ethical concerns, data privacy issues, and the potential societal impact of widespread automation. Furthermore, the implementation of AI technology is often accompanied by adoption risks, including the possibility of integration failures and the emergence of new risks, such as cybersecurity vulnerabilities.

These uncertainties make it essential for investors to have access to high-quality information regarding firms' AI initiatives and the associated risks. The current disclosure regulatory framework, however, lacks explicit guidelines or standards for AI reporting, making such disclosures voluntary. The SEC has raised significant concerns about the credibility of AI-related disclosures, warning against the practice of "AI washing," where firms overstate their AI capabilities to appear more advanced than they are.[2] Investors share these concerns, becoming

[1] https://www.kornferry.com/institute/humans-still-wanted-the-future-of-work-in-an-ai-driven-world.

[2] SEC Chair Gary Gensler, in a September 2024 statement, emphasized the need for companies to ensure that their AI-related claims align with their actual capabilities, signaling the regulator's increasing focus on this issue. See https://www.thecorporatecounsel.net/blog/2024/09/sec-chair-addresses-ai-washing-by-public-companies.html.

increasingly wary of exaggerated claims, which have contributed to a rise in the number of securities class action lawsuits.[3] Against this backdrop, there is limited evidence on voluntary AI disclosure's determinants and information content, presenting substantial challenges for external stakeholders seeking to assess corporate AI initiatives and engagements. The purpose of this paper is to fill in this important gap by providing a systematic analysis of corporate voluntary AI disclosure in their annual reports.

Specifically, we ask three questions regarding corporate voluntary AI disclosure. First, to what extent do publicly traded companies voluntarily disclose AI-related information in annual reports, and how have these disclosures evolved over time and across industries? Second, what are the determinants of AI disclosure? Are companies using AI disclosure merely to appease investor interests without substantial involvement in AI, or do these disclosures reflect genuine engagement with AI initiatives? Third, what is the information content of AI disclosures? Do they contain forward-looking material information about a firm's future performance prospects? We believe answering these questions is the first important step in understanding the implications of corporate voluntary AI disclosure.

Addressing these questions is challenging due to the voluntary nature of AI disclosures in annual reports and the lack of standardized reporting frameworks. AI-related information is often dispersed throughout firms' annual reports and may lack a consistent format, making it difficult to accurately and consistently measure AI disclosure across firms. To address this challenge, we develop a two-step procedure that integrates traditional keyword-based textual analysis with the latest large language models (LLMs).

[3] AI disclosure-related lawsuit filings are increasing in recent years as reported by the Stanford Law School Securities Class Action Clearinghouse: https://securities.stanford.edu/current-trends.html#collapse1. For instance, a recent case was filed against GitLab, alleging that the company misled investors about its AI capabilities and their expected impact on market demand: https://www.dandodiary.com/wp-content/uploads/sites/893/2024/09/GitLab-complaint-2.pdf.

In the first step, we utilize a keyword-based textual analysis approach to identify AI-related disclosures within U.S. public firms' 10-K filings (obtained from SEC EDGAR). We start with the keyword "artificial intelligence" to ensure clarity in identifying AI disclosure. Subsequently, we expand the keyword list by manually reviewing 10-K statements and incorporating frequently mentioned related words and bigrams until the accuracy of algorithm-identified AI disclosure instances reaches 99% compared with human reading outcomes in randomly selected 10-K samples.

While the keyword-based method facilitates the initial identification of AI-related text segments, it has two notable limitations. First, focusing solely on sentences containing specific keywords may overlook relevant context. AI-related discussions might occur in the surrounding text or be implied rather than explicitly stated, leading to inaccurate assessments of a firm's AI usage based purely on keyword counts. Second, traditional textual analysis methods struggle to capture the nuanced context within the identified sentences. This limitation makes it challenging to determine the specific nature of AI usage being discussed. For instance, it is difficult to discern whether the mention of AI pertains to revenue-generation activities, such as product development and pricing optimization, or cost-reduction activities, such as inventory management or operational efficiency improvement.

To overcome these limitations, we incorporate advanced large language models (LLMs), specifically ChatGPT, in the second step of our analysis. For each AI-related keyword identified in the first step, we extract an 800-word window surrounding the keyword and prompt ChatGPT to assess the relevance of the information to AI usage. ChatGPT then classifies the types of AI usage mentioned into five categories: Product Development (leveraging AI to design or enhance new products), Pricing Optimization (using AI to set or adjust prices dynamically based on market

conditions), AI Product Provider (offering AI-based solutions to customers as part of the firm's product portfolio), Inventory Management (utilizing AI to forecast demand and optimize stock levels), and Operational Efficiency (applying AI to streamline and automate internal processes). Integrating ChatGPT's contextual understanding provides a more accurate and granular perspective on AI activities.

Using these new measures of AI disclosure, we first document patterns associated with voluntary AI disclosure in the 10-K statements of U.S. public firms between 2010 and 2023. We observe that only 2.36% of U.S. public firms disclosed AI-related matters in 2010, a figure that surged to 20.02% in 2023. This rapid growth in AI disclosure over the past decade is pervasive across all industry sectors. When breaking down 10-K sections, we find that AI disclosures are concentrated in the business description (Item 1), risk factors (Item 1A), and management discussion and analysis (Item 7), reflecting the positioning of AI within the firm's operational framework, strategic considerations, and associated risks and opportunities.

One crucial question is whether this surge in AI disclosures represents a superficial response to investor interest or a genuine reflection of firms' commitment to AI initiatives. To shed light on this question, we examine the association between AI disclosures and firms' actual AI engagement. Following Babina et al. (2024), we proxy for AI engagement using the proportion of employees possessing AI-related skills (identified in their resumes). Firms typically require specialized AI expertise—such as algorithm design, data analytics, and machine learning model development—to implement AI initiatives effectively. Thus, the proportion of AI-skilled employees captures firms' tangible investments in AI activities.

Our results reveal that firms with higher fractions of AI-skilled employees are more likely to provide voluntary AI disclosures in their annual reports. Specifically, a one-percentage-point

increase in the fraction of AI-skilled employees corresponds to a 0.6 percentage point increase in the probability of AI disclosure. The results indicate that voluntary AI disclosures are not merely cosmetic attempts to align with current trends; rather, they reflect substantive firm engagement in AI-related activities. Moreover, larger firms, firms with higher market valuations, and younger firms are more inclined to issue AI disclosures, suggesting that resource availability, competitive pressures, and an innovative organizational culture may shape the propensity to disclose AI activities.

Effective corporate disclosure should not only reflect current activities but also provide forward-looking insights that inform investors about potential risks and opportunities (Merkley, 2014; Noh, So, and Weber, 2019; Aghamolla and An, 2021; Kremer, Schreiber, and Skrzypacz, 2024). To shed light on the information content of AI disclosures, we extend Babina et al. (2024) by comparing the power of AI disclosures to that of the AI employee fraction in predicting firm growth and innovations. We find that current-year AI disclosures are positively associated with subsequent-year sales growth, employment growth, capital investment, and R&D input. Additionally, current-year AI disclosures are negatively associated with subsequent-year operational inefficiency as measured by the cost of goods sold (COGS) relative to sales and operating expenses relative to the number of employees. The predictability of AI disclosure persists even after controlling for the fraction of AI-skilled employees, suggesting that AI disclosures contain incremental, forward-looking information related to both opportunities and risks.

To gain further insight into the specific activities underlying these disclosures, we leverage ChatGPT's classification of AI usage into five categories. We group Product Development, Pricing Optimization, and AI Product Provision as revenue-generating activities and Inventory

Management and Operational Efficiency as cost-reducing activities. We observe that the number of firms employing AI for both revenue generation and cost reduction has grown steadily over time. Moreover, both sets of activities contribute to explaining future growth and operational efficiency, highlighting the dual importance of AI in shaping firm performance.

In addition to informing future cash flows through revenue and cost-related factors, AI disclosures may also offer valuable information about the firm's risk profile. We use keyword searching and GPT-based analysis to classify AI-related risks disclosed in annual reports' risk factors (Item 1A) sections into six categories: regulatory, operational, competitive, cybersecurity, ethical, and third-party risks. We find that AI risk disclosures correlate with higher levels of tail risk, as measured by stock and option-implied volatility. Among the six risk categories, regulatory, competitive, and ethical concerns have the strongest associations with investor perceptions of increased risk.

Our study contributes to the growing literature on measuring firm-level AI engagement and its consequences. Eisfeldt, Schubert, and Zhang (2023) find a positive equity premium for firms whose employees' occupations having a greater exposure to generative AI. Babina et al. (2024) use job posting information to measure a firm's fraction of AI-related employees and find it is positively associated with a firm's growth and innovations. Acemoglu, Autor, Hazell, and Restrepo (2022) use online job postings to measure occupational AI exposures and find that when AI-exposed establishments adopt the technology, hiring for non-AI positions decreases, and the skill requirements in the remaining roles shift accordingly. Our study proposes a new measure of a firm's AI disclosure, which is significantly related to a firm's present engagement in AI activities yet includes additional predictive power on firm growth, innovation, and operation efficiency. This

new measure complements other firm-level AI engagement metrics and offers a new approach to analyzing a firm's AI activities.

Our study contributes to the growing literature on extracting information from corporate voluntary disclosures. Li, Lundholm, and Minnis (2013) use annual reports to measure competition, while Kim, Wang, and Wu (2023) and Kölbel, Leippold, Rillaerts, and Wang (2024) examine climate risk exposure in the risk factor sections of 10-K statements. Sautner, van Lent, Vilkov, and Zhang (2023), along with Li, Shan, Tang, and Yao (2024), extract firm-level climate risk measures from earnings call transcripts, and Bourveau et al. (2022) and Zhang (2022) analyze human capital disclosures in annual reports. Additionally, Hassan, Hollander, van Lent, and Tahoun (2019) provide firm-level political risk measures using textual analysis of quarterly earnings calls. By examining AI-related information in annual reports, our research introduces a novel dimension to corporate disclosure analysis. Specifically, our study of AI activities offers deeper insights into firms' engagement with AI, thus contributing to a broader understanding of the evolving corporate disclosure landscape and the role of advanced technologies in shaping corporate strategies.

Finally, our study offers policy implications, particularly in light of increasing regulatory efforts in artificial intelligence. President Biden's Executive Order on AI in October 2023 underscores the importance of developing standards and practices to enhance consumer safety and privacy.[4] By analyzing AI disclosure through textual analysis in annual reports, regulators can gain valuable insights into firms' AI strategies, potential risks, and approaches to safeguarding consumer interests. This information can inform the development of regulatory frameworks tailored to the rapidly evolving AI landscape in corporate settings.

[4] https://www.whitehouse.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/.

2. Measuring AI Disclosure in SEC Filings

2.1. AI-related keywords identification

Our first task is to compile a list of AI-related keywords that allow us to pinpoint potential AI disclosure. We obtain 10-K filings from 2010 to 2023 using Stage One 10-X Parse Files from the Notre Dame Software Repository for Accounting and Finance, established by Loughran and McDonald (2016). In the first step, we identify all 10-Ks that mention "artificial intelligence."[5] After this pre-selection, we randomly choose 25 reports with "artificial intelligence" mentioned in the 10-Ks for each year from 2010 to 2023 (350 reports in total). We manually read through this set of 10-Ks and create the initial AI-related keyword list by including the frequently mentioned AI-related terminology words and bigrams in these reports. To ensure the high relevance of keywords, we limit the initial list to 25 keywords.

In the next step, we improve this initial AI-related keyword list using an out-of-sample learning approach. The out-of-sample learning approach compares classification outcomes using the keyword list approach versus the manual review approach for 50 randomly selected firms yearly for the 14-year sample period (700 reports). A report is automatically classified as AI-related if it includes words in the AI-related keyword list. Then, we use human review to classify if a report is AI-related. The accuracy of using the AI-related keyword list in classification is measured as the percentage of automatically identified AI-related reports by keyword list that are consistent with manually identified AI-related reports. If accuracy is less than 98% (more than one misclassified report), we read manually identified AI-related reports that differ from keyword list classification. We then add new keywords in each improvement round. We repeat this improvement process and

[5] Sample 10-K report paragraphs with "artificial intelligence" explicitly referred to are shown in Appendix A1.

stop once the accuracy ratio reaches 98%. The final AI disclosure keyword list is presented in Table A2, with the frequency of the keyword showing in all 10-K filings and AI-disclosing ones.

Table A3 tracks the top 3 most frequently mentioned AI keywords in the final list from 2010 to 2023. In the early years (2010 to 2013) within our sample, the AI keywords with the highest appearance frequency in the 10-K reports are "business intelligence," "AI/ML," and "data mining." In the following years (2014 to 2017), "big data" picks up the trend and becomes the top 3 highest referenced AI keywords, while "business intelligence" remains the highest cited AI keyword. Starting in 2018, "artificial intelligence" and "machine learning" are the top two highest-mentioned AI keywords in the corporate annual reports. The keyword "business intelligence" loses its third position and is replaced by "data science" in 2022. The evolution of AI technologies over time is also reflected in the first appearance of keywords in the annual reports. For example, "big data" first showed up in 2011, suggesting the focus of AI applications shifted to data science. A more recent AI advancement, "deep learning," appeared in the 2014 annual reports. In 2023, the reference to the "large language model" quickly occupies the reports, coinciding with the introduction of ChatGPT in late 2022.

2.2. AI disclosure identification and classification

The keyword list approach has a few potential limitations. First, it is susceptible to human bias and inconsistency in classifying 10-K reports. Second, the same keywords can have different meanings in varying contexts. For example, "data mining" might refer to applications of AI or machine learning algorithms, but it could also be irrelevant to AI when discussing data processes related to analyzing customer behavior. Third, using keywords alone makes it challenging to accurately identify the purpose of a firm's engagement with AI. To address these issues, we utilize

the latest large language model, ChatGPT-4o, to improve the precision of classifying AI-related disclosures and identify the motivations behind a firm's AI initiatives.

We first apply the aforementioned keyword list to identify potential AI disclosures in the 10-K filings. Since we focus on AI applications, we search for AI-related keywords in Item 1 (Business) or Item 7 (Management's Discussion and Analysis). When a keyword is identified, we extract an 800-word window around it (400 words before and after) and employ the ChatGPT-4o model to review this text.[6] The model then determines whether the disclosure pertains to AI (*AI Disclosure*) and, if so, classifies it into one of five categories: Product Development, Pricing Optimization, AI Product Provider, Inventory Management, and Operational Efficiency.

We further classify Product Development, Pricing Optimization, and AI Product Provider as revenue-related AI disclosures (*AI Disclosure-Revenue*), while Inventory Management and Operational Efficiency fall under cost-related AI disclosures (*AI Disclosure-Cost*). For this classification task, we apply a "zero-shot prompting" approach due to the challenge of providing a solid and comprehensive definition of AI. Additionally, we require ChatGPT to provide a probability score for the AI-related classification and for each application category separately. The model also provides an explanation for these scores, allowing an audit of its reasoning. Table A4 details the GPT prompts used for classification and categorization. This procedure results in 6,028 firm-year observations with AI disclosures from 1,985 unique firms.

Similarly, using a combination of keyword searching and GPT-based analysis, we classify AI-related risks in Item 1A into six key categories. Regulatory risks involve compliance with current and emerging government regulations, while operational risks stem from system malfunctions or disruptions. Competitive risks arise from the threat of rivals leveraging superior AI technologies,

[6] In most cases, the length of the extracted corpus exceeds 800 words to ensure the sentences remain complete.

and cybersecurity risks pertain to the potential exploitation of AI vulnerabilities by malicious actors. Ethical risks focus on fairness, transparency, and potential public backlash due to biased AI decisions. Lastly, third-party risks are linked to reliance on external AI vendors, which could result in data privacy issues or supply chain disruptions. Table A.6 details the GPT prompts used for classification and categorization. This procedure results in 2092 firm-year observations with AI disclosures from 796 unique firms.

2.3. Measuring AI intensity

To measure AI intensity in disclosures, we utilize ChatGPT to identify AI-related content within the 800-word corpus. Our identification process employs the "few-shot prompting" strategy, which generates clear and accurate responses when the task is well-defined. This strategy involves three steps: 1) System Prompt Creation: We create a system prompt that outlines the setup of the task. 2) Example User-Assist Prompts: We develop example prompts to illustrate the expected output. In these examples, the user prompt asks ChatGPT to remove content unrelated to AI disclosure, while the assist prompt provides the sample answers. 3) Task Execution: For each task, we provide the GPT model with the system prompt, the example user-assist prompts, and a new user prompt that includes the AI disclosure text extracted from the 10-K filings. The model then produces responses similar to the assist prompt as instructed.

Table A5 details our prompts and classification procedure. *AI intensity* is measured as the number of words in AI disclosures processed by GPT, scaled by the total number of words in the corpus extracted from 10-K filings. Similar to the AI disclosure indicator, we also create two measures, *AI Intensity-Revenue,* and *AI Intensity-Cost*, to capture AI disclosure intensity in these two dimensions.

3. Data and Summary Statistics

We merge AI disclosure data with multiple datasets to analyze AI disclosure's determinants and information content. Firm characteristics and financial information are from the Compustat database. Information on stock performance is from CSRP. Finally, we obtain institutional ownership from Thomson Reuters 13F. Combining these databases creates a final sample containing 4,290 unique firms from 2010 to 2023, with about 32,000 firm-year observations.

The summary statistics of the main variables used in this study are reported in Table 2, with variable definitions provided in Table A6. The mean of AI Disclosure is 9.4%, indicating that 9.4% of 10-K statements during the sample period disclose AI information. AI Disclosure-Revenue (Cost) has an average value of 5.9% (4.5%), suggesting that 5.9% (4.5%) of 10-K filings disclose revenue (cost)-related AI information. The AI risk disclosure is relatively sparse compared with the AI usage information. 1.2% of the sample contains the AI risk disclosure in item 1A of the 10-K statement. Another key variable in this study is the AI Employee Share, developed by Babina et al. (2024), representing the proportion of employees with AI-related skills (as identified in their resumes) relative to the total workforce. On average, our sample firms have 0.1% of employees with AI-specific knowledge, 3,300 employees on average, $404.7 million in gross sales, $849.8 million in total assets, cash holdings at 22.4% of total assets, a market-to-book ratio of about 2.2, and an average of 13.8 years since IPO. Table 3 shows the pairwise correlation between the main variables. It shows that the likelihood of having AI disclosure is positively correlated with the share of employees with AI knowledge.

[Insert Tables 2 and 3 about here]

4. Trends and Patterns in AI Disclosure

Table 1 reports the number of AI-disclosing firms each year in the sample, demonstrating a rapid growth in AI disclosure among U.S. public firms over the past decade. In 2010, only 83 firms (2.36% of firms) disclosed AI-related issues in their 10-K statements. This number surged to 834 firms (20.02%) in 2023. Figure 1 illustrates this trend, clearly revealing an accelerated growth regime post-2017. This pattern aligns with the significant advancements in AI technologies in 2017 and 2018, driven by breakthroughs in deep learning, particularly in natural language processing and image recognition tasks.[7]

[Insert Figure 1 and Table 1 about here]

We calculated the proportions of AI disclosure firms within each industry sector using the two-digit NAICS codes to investigate the pattern of AI disclosure across different industries. Figure 2 illustrates the evolution of AI disclosure in various industries from 2010 to 2023. Two main observations emerge. First, AI disclosure has become ubiquitous, with all industry sectors reporting AI disclosures by 2023. This was not the case in 2010 when many industries, such as healthcare, construction, utilities, mining, and agriculture, had no AI disclosures. In addition, in 2023, over 30% of firms in sectors like information, education, professional and business services, agriculture, administrative support, and healthcare reported AI disclosures in their 10-K statements. Clearly, the impact of AI technology has permeated the entire economy.

[Insert Figure 2 about here]

[7] Notable contributions include the introduction of the transformer architecture in "Attention is All You Need" by Vaswani et al. (2017), which revolutionized natural language processing, and the development of BERT, a groundbreaking language representation model, as outlined in "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" by Devlin et al. (2018). These years also saw increased adoption of AI across various industries and heightened discussions surrounding AI ethics, regulation, and the development of autonomous systems, as documented in the “2018 AI Index Report” by Stanford University.

Second, industry sectors with the highest AI disclosure have shifted over time. Panel A of Figure 2 shows that, in 2010, the three industries with the most intensive AI disclosure were professional and business services (11.18%), information (11.11%), and administrative and entertainment (9.52%). By 2023, the top three industries had shifted to information (56.47%), education (52.94%)[8], and professional and business services (49.23%).[9] The high percentage of firms in the education industry discussing AI in their annual reports is somewhat surprising. This underscores how AI has evolved into a highly sophisticated educational technology, presenting both significant challenges and opportunities for the education sector.

Panels B and C of Figure 2 break down AI disclosure into revenue and cost categories. We find that revenue and cost disclosure follow a similar distribution to overall disclosure, with the top three industries in 2023 being information (42.46%), education (35.29%), and professional and business services (29.23%) for revenue, while the top three industries for cost disclosure are education (29.41%), information (27.59%), and professional and business services (26.92%).

In Panel D of Figure 2, the growth of AI risk disclosure is significant among different industries. In 2010, rare industrial sectors have the AI risk disclosure. In 2023, the top three industrial sectors with AI risk disclosure are information (22.20%), administrative support (18.42%), and education

[8] We take a closer look at AI reporting in the education sector and read all the 2023 10-K reports from the education sector. The annual reports show that educational institutions actively use artificial intelligence and machine learning to provide learning resources to their customers. For example, Nerdy Inc. stated in the 10-K: "Our purpose-built proprietary platform leverages technology, including *artificial intelligence* (AI), to connect students, users, parents, guardians, and purchasers…". Coursera Inc. included the following in the 10-K statement: "Coursera's data and *machine learning* systems drive personalized learning, and skills benchmarking."

[9] This is partially due to the number of firms in this industry sector being relatively small (15 firms in 2023 in our sample). We also investigate the industry sectors (by 2-digit NAICS code) with the most companies disclosing AI-related matters in 2010 and 2023. The top 3 industry sectors with the most firms disclosing AI were manufacturing, information, and professional and business services in 2010. The top 3 industry sectors with the most firms disclosing AI switch to information, manufacturing, and finance/insurance in 2023. The number of AI disclosing firms may be biased by the total number of firms in that industry but represents the industry-wise AI disclosure intensity.

(17.65). Almost all of the industrial sectors have firms with AI risk disclosure (except for agriculture, entertainment, and other services).

5. Determinants of Corporate AI Disclosure

The public generally perceives AI technologies as a novel component that could provide firms with a significant competitive edge by enhancing operation efficiency, decision-making, and innovation. Firms may choose to disclose their AI initiatives as a strategic move to showcase their commitment to innovation and technological advancement, thereby enhancing investor confidence and attracting potential stakeholders. However, it is also possible that some firms may disclose AI initiatives to attract investors, even if they lack substantial AI activities, as a means of leveraging investor interest in AI technologies. Therefore, the voluntary nature of AI disclosure raises the question of what factors influence the firm to disclose, particularly whether such disclosure is related to a firm's actual engagement in AI.

To gain insights into the determinants of AI disclosure, we investigate how a firm's AI disclosure is related to its AI activities and other key firm characteristics by estimating the following regression:

$$AI\ Disclosure_{i,t+1}/AI\ Intensity_{i,t+1} = \beta \cdot AI\ Employee\ Share_{i,t} + \lambda \cdot Other\ Firm\ Chacteristics_{i,t} + D_j + D_t + \epsilon_{i,t+1}$$

where the dependent variable is either *AI Disclosure* or *AI Intensity*. We use the logistic regression models when the dependent variable is *AI Disclosure* or *AI Disclosure-Revenue* (*Cost*). When the dependent variable is *AI Intensity* or *AI Intensity-Revenue (Cost)*, we use OLS regression. The independent variable $AI\ Employee\ Share_{i,t}$ is the fraction of AI employees developed by Babina

et al. (2024), gauging a firm's real activities in AI. The advancement of AI necessitates the recruitment of employees proficient in AI, as they hold specialized expertise crucial for tasks like algorithm design, data analysis, and model development. Consequently, the proportion of AI-skilled employees within a company signifies the resources dedicated by firms to AI endeavors. Other firm characteristics include firm size (logarithm of total assets), return on assets, cash holdings, leverage ratio, net PP&E, market-to-book ratio, and (logarithm of) firm age.

The estimation results are shown in Table 4. Columns (1), (2), and (3) report the results when the dummy *AI Disclosure* is the dependent variable. In Column (1), the coefficient on $AI\ Employee\ Share_{i,t}$ is positive and significant at a 1% level. It shows that the share of employees with AI skills positively affects the likelihood of AI disclosure. Economically, one percentage increase in AI Employee Share leads to a 0.6% increase in the likelihood of AI disclosure, which represents a 6.4% increase relative to the sample mean (0.094).[10] The results show that a firm's real engagement in AI is a key factor affecting AI disclosure.

The results also show that firms with larger sizes, larger cash holdings, lower leverage, lower tangibility, higher growth opportunities, and younger ages are more likely to have AI disclosures. Larger firms are more likely to engage in AI disclosure due to their extensive resources and capabilities to invest in cutting-edge technologies like AI, signaling their commitment to innovation and maintaining market leadership. Cash holdings and leverage are indicative of a firm's financial resources and associated risks. The coefficients show that firms with greater financial capability and lower risk profiles are more likely to disclose AI, which suggests that AI projects are resource-intensive and risky. Lower tangibility firms, relying more on intangible assets like technology, might use AI disclosure to signal their future growth prospects. Similarly, firms

[10] When calculating the marginal effect, we assume the rest of the covariates taking the sample mean value.

with higher growth opportunities and younger age might prioritize innovation to capitalize on growth potential. Thus, disclosing their AI initiatives serves to signal their commitment to technological advancement, aligning with their growth-oriented strategies.

Columns (2) and (3) report the determining factors of revenue-related and cost-related AI disclosures, respectively. The coefficients in column (2) remain largely unchanged, except that ROA is negatively correlated with revenue-related AI disclosure. This suggests that companies tend to disclose more AI information related to boosting revenue during periods of lower performance. In column (3), the coefficient for AI Employee Share has a lower magnitude and loses significance, indicating that most firms hire AI employees primarily to enhance revenue rather than reduce costs. Columns (4), (5), and (6) present the results for AI Intensity measures. The coefficients for AI Employee Share and other characteristics are largely similar to those for AI Disclosure. Overall, the results indicate that AI disclosure extends beyond merely catering to investors; it is intricately linked to a firm's engagement with AI technologies.

6. Information Content of AI Disclosure

Corporate disclosure should not only reflect the present state of activities but also encompass forward-looking information pertaining to both opportunities and risks (Merkley, 2014; Noh, So, and Weber, 2019; Aghamolla and An, 2021; Kremer, Schreiber, and Skrzypacz, 2024). However, a firm's AI voluntary disclosure may not necessarily include material information useful for investors to predict its future performance. Some firms may disclose comprehensive details about their AI initiatives, including strategies, investments, risks, and outcomes, providing investors with valuable insights into the firm's prospects. Conversely, other firms may opt for minimal disclosure, withholding sensitive information to maintain a competitive advantage or mitigate perceived risks.

Additionally, the complexity and novelty of AI technologies present challenges in accurately quantifying their potential impact on future performance. Therefore, even if a firm provides AI disclosure related to its present AI activity, it might not include adequate information that is forward-looking of a firm's future performance. In this section, we assess the information content of AI disclosure and investigate if AI disclosure is related to a firm's future growth and corporate costs.

6.1. AI disclosure and firm growth

We estimate the following regression to investigate the information content of AI disclosure about a firm's future growth and innovation.

$$Firm\ Growth_{i,t+1} = \beta_1 \cdot AI\ Disclosure_{i,t} + \beta_2 \cdot AI\ Employee\ Share_{i,t} + \lambda \cdot Controls_{i,t} + D_j + D_t + \epsilon_{i,t+1}.$$

where the dependent variable $Firm\ Growth_{i,t+1}$ is a firm's performance in year t+1. Similar to Babina et al. (2024), we use three different dependent variables to measure a firm's growth, including the logarithm of the number of employees, the logarithm of sales and investment. $AI\ Disclosure_{i,t}$ and $AI\ Employee\ Share_{i,t}$ are two main independent variables. Both variables are measured in year t, allowing us to examine its predictability of performance in year t+1. Other control variables include the logarithm of total assets, return on assets, cash holdings scaled by total assets, leverage ratio, net PP&E scaled by total assets, market-to-book ratio, and the logarithm of firm age. D_j and D_t represent the industry sector (2-digit NAICS code) fixed effect and year fixed effect.

Columns (1) and (2) in Panel A of Table 5 report the results with the logarithm of employment as the dependent variable. The coefficient on $AI\ Disclosure_{i,t}$ is positive and statistically

significant, suggesting that AI disclosure is associated with increased total employees. Economically, in column (2), the coefficient on $AI\ Disclosure_{i,t}$ is 0.139 (t-stats. = 6.653), suggesting that firms with AI disclosure are associated with a 13.9% increase in employee population compared to the firms without AI disclosure. These results demonstrate that AI disclosure positively predicts the growth in the number of employees within a firm, whereas the AI employee share does not have a similar effect. This evidence implies that AI disclosure provides additional explanatory information regarding firms' employment and staffing.

[Insert Table 5 about here]

Columns (3) and (4) present the results with the logarithm of sales in year t+1 as the dependent variable. The coefficients for AI disclosure are consistently positive and significant across both regression specifications, with a coefficient of 0.16. Economically, firms with AI disclosure experience a 16% increase in gross sales in the following year. Columns (5) and (6) of Table 5 report the results with firm investment as the dependent variable. When all control variables are included in column (6), the coefficient for AI disclosure is 0.002 (t-stats. = 3.588). This coefficient is also economically significant, indicating that AI-disclosing firms tend to have 6.1% higher capital expenditures in the next year relative to the sample mean (0.033) than firms without AI disclosure. Columns (7) and (8) show the impact of AI disclosure on R&D expenditures in year t+1. With all control variables included in column (8), the coefficient for AI disclosure is 0.464 (t-statistic: 11.371), suggesting that firms disclosing AI information tend to have 15.3% higher capital expenditures compared to non-disclosing firms relative to the sample mean (2.806).

Panel B of Table 5 presents the results by separating AI Disclosure into revenue and cost components. In columns (1), (4), (7), and (10), we include only *AI Disclosure-Revenue*, while in columns (2), (5), (8), and (11), we only retain *AI Disclosure-Cost.* In columns (3), (6), (9), and

(12), we put both into the regressions. The coefficients of *AI Disclosure-Revenue* and *AI Disclosure-Cost* are mainly positive and statistically significant across all columns, but the coefficient of *AI Disclosure-Revenue* is insignificant in *log(Employees), log(Sales),* and *CAPX/Assets.*

Panel C reports the results from estimating the predictability of AI intensity variables on firm growth. Columns (1), (4), (7), and (10) present the results for overall AI intensity. Columns (2), (5), (8), and (11) show the results for revenue-related AI intensity, while columns (3), (6), (9), and (12) present the results for cost-related AI intensity. We find that firms with higher AI intensity in SEC filings tend to experience increases in the number of employees, sales, investment in capital expenditures, and R&D. However, firms disclosing more intensive revenue-related AI information are only associated with higher R&D in the following year. In contrast, firms with more intensive cost-related AI information are more likely to have a higher number of employees, increased sales, and larger investments in capital expenditures and R&D.

6.2. AI disclosure and costs

A key function of AI disclosure is to report a firm's commitment to reducing costs and enhancing efficiency. In this section, we investigate whether AI disclosure is indeed followed up by cost reduction. We focus on the cost of goods sold (COGS) and operating expenses, as these constitute a substantial portion of corporate costs and are closely related to AI applications. We scale COGS by total sales (*COGS/Sales*) and number of employees (*COGS/Employees*). For operating expenses, we scale by the number of employees (*Operating Expense/Employees*).

Panel A of Table 6 reports the regression results estimating the predictability AI disclosure on corporate costs. Columns (1) - (4) use COGS as the dependent variable, with COGS scaled by sales in columns (1) and (2), and by the number of employees in columns (3) and (4). The coefficient of AI Disclosure indicates that AI disclosure is negatively associated with next year's

COGS. Economically, firms with AI disclosure in SEC filings experience a 23.04% (14.28%) decrease in COGS scaled by sales (number of employees) in the following year. Additionally, AI disclosure is associated with a 13.36% drop in *Operating Expenses/Employees*. Panel B presents the results by separating AI disclosure into revenue-related and cost-related components. Both *AI Disclosure-Revenue* and *AI Disclosure-Cost* are associated with a significant decline in corporate costs in the subsequent year across all columns. Panel C shows the results for AI intensity variables. Columns (1), (4), and (7) use overall AI Intensity. Columns (2), (5), and (8) use revenue-related AI intensity, while columns (3), (6), and (9) use cost-related AI intensity. We find that all AI intensity variables are negatively correlated with corporate costs for all proxies, including COGS/Sales, COGS/Employees, and Operating Expenses/Employees. Overall, our findings suggest that AI disclosure and AI intensity signal firms' commitment to reducing corporate costs.

[Insert Table 6 about here]

6.3. AI risk disclosure and firm risk

The risk factor disclosure in item 1A of the 10-K statement provides the forward-looking risk assessment of the underlying business from the firm managers. We consider the AI risk-related disclosure to have distinguished effects on the financial consequences of the firm. Given the forward-looking risk-related nature in item 1A disclosure, we estimate the following regression to investigate the information content of AI risk disclosure about a firm's risk profile.

$$Firm\ Risk_{i,t+1} = \beta_1 \cdot AI\ Risk\ Disclosure_{i,t} + \lambda \cdot Risk\ Controls_{i,t} + D_j + D_t + \epsilon_{i,t+1}.$$

where the dependent variable $Firm\ Risk_{i,t+1}$ is a firm's risk related proxies in year t+1. We assess the firm risk profile using the stock inferred and option implied information. We select the idiosyncratic volatility estimated from the annualized residual volatility from the Fama and French (1993) three factor model to represent the stock volatility risk. We follow the option implied

volatility literature (Bakshi, Kapadia, and Madan, 2003; Ilhan, Sautner, and Vilkov, 2021; Kelly, Pástor, and Veronesi, 2016) and construct the slopeness of option-implied volatility on option delta (SlopeD), the variance risk premium (VRP), the model-free option implied volatility (MFIV), and the model-free option implied skewness (MFIS). The option-implied risk metrics reflect the forward-looking firm tail risk and volatility risk.

As for the control variable, we include the logarithm of total assets, the logarithm of firm age, Tobin's Q, book leverage ratio, the cash holding scaled by the total assets, intangible assets scaled by the total assets, the stock return sensitivity with respect to the market return, the operating expense scaled by the total assets. The control variables cover the firm characteristics that may impact the firm's risk profile. The main independent variable is the AI risk disclosure dummy variable. The coefficient of the variable estimates the incremental contribution of the AI risk disclosure on the firm risk.

Using the firm level stock and option implied risk measures as the main dependent variable, we perform the regression analysis with the AI risk disclosure quantified from section item 1A. Panel A of Table 7 presents the regression results using the aggregate level AI risk disclosure dummy variable as the main independent variable. We find that the AI risk disclosure in the 10-K statement predicts a higher level of firm tail risk. The coefficient of the AI risk disclosure variable is statistically and economically significant for stock inferred and option implied volatility measures. This indicates that the AI related risk factors are priced in the stocks and options of the firms. The investors gain incremental risk assessment and react to the corporate AI risk disclosure.

[Insert Table 7 about here]

Additionally, we decompose the topics of AI risk disclosure into six categories: regulation risk, operational risk, competition risk, cybersecurity risk, ethical risk, and third-party risk. We rerun the firm risk regressions using the categorical AI risk disclosure dummy variable as the main independent variable. The empirical results are summarized in Panel B of Table 7. Each line in the table represents a standalone regression, and we report only the coefficients of the main independent variables (categorical AI risk disclosure) for brevity. Overall, the categorical AI risk disclosure forecasts higher firm tail risk. Among the magnitude of the categorical AI risk disclosure variables, regulation risk, competition risk, and ethical risk seem to be the major AI-related risk factors that raise investors' concerns about the stock and option markets.

In summary, AI risk-related disclosures in Item 1A of the 10-K statement provide essential forward-looking insights into a firm's risk profile. Our findings indicate that investors integrate these disclosures into their risk assessments, leading to their reflection in stock and option pricing. These results emphasize the pivotal role of AI-related risk disclosures in shaping market perceptions and underscore the importance for firms to carefully evaluate the financial implications of their AI-related risk communication.

7. Conclusion

AI has emerged as the most transformative technology in today's business world. AI offers significant potential to enhance future cash flows by enabling innovative products, more precise customer targeting, and operational efficiency through automation and optimization of supply chains and production processes. However, despite these benefits, AI adoption brings substantial uncertainty to investors, arising from evolving regulations, concerns about privacy and ethics, and the potential for cybersecurity vulnerabilities. Additionally, the success of AI implementation is

not guaranteed, as it involves risks of integration failure and unforeseen consequences. Given these uncertainties, investors require high-quality information about firms' AI initiatives and the associated risks. Yet, the current regulatory framework lacks specific guidelines for AI reporting, making disclosures largely voluntary. This paper takes the first step in analyzing voluntary AI disclosure.

We use a combination of traditional textual analysis techniques and the latest LLMs like ChatGPT to measure and analyze AI disclosure in corporate annual reports. In the first step, keyword-based analysis is used to identify AI-related information in firms' 10-K filings. However, traditional textual analysis methods have limitations in capturing the full context and nature of AI usage. To address this, we use ChatGPT to classify AI-related disclosures into five categories: product development, pricing optimization, AI product provision, inventory management, and operational efficiency. This approach allows for a deeper understanding of how firms engage with AI, grouping the first three categories as revenue generation activities and the latter two as cost reduction activities.

Our findings reveal that the number of firms adopting AI for both revenue generation and cost reduction has grown substantially since 2010. Furthermore, both revenue generation and cost reduction through AI are associated with future growth and operational efficiency, indicating that AI plays a crucial role in shaping firm performance. However, the disclosure of AI-related risks in 10-K filings is equally critical for understanding the broader implications of AI adoption. AI risk-related disclosures in Item 1A of 10-K filings provide forward-looking insights into potential challenges and uncertainties faced by firms adopting AI technologies. These risks include regulatory compliance, ethical concerns, cybersecurity vulnerabilities, competition, and third-party dependencies, among others. Our analysis highlights that AI risk disclosures have significant

implications for firm risk profiles. Investors respond to these disclosures by pricing them into financial instruments such as stocks and options. Moreover, AI risk factors like regulatory, competition, and ethical risks stand out as major concerns for investors. These findings underscore the importance of not only highlighting AI opportunities but also providing comprehensive risk-related disclosures to enable investors to make informed assessments about firms' AI strategies. Overall, this study underscores the importance of accurate AI disclosures for investors to assess current engagements and future opportunities and risks associated with AI adoption.

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Figure 1: Trends of AI Disclosure

The graph illustrates the fraction of firms with AI disclosure over the total number of firms. The solid line plots the trend of AI disclosure shown in the full 10-K filings. The dot-dashed line plots the trend of AI disclosure in section Item 1 business. The dotted line plots the trend of AI disclosure in section Item 7, management discussion and analysis. The dashed line plots the trend of the AI risk disclosure in Item 1A risk factors.

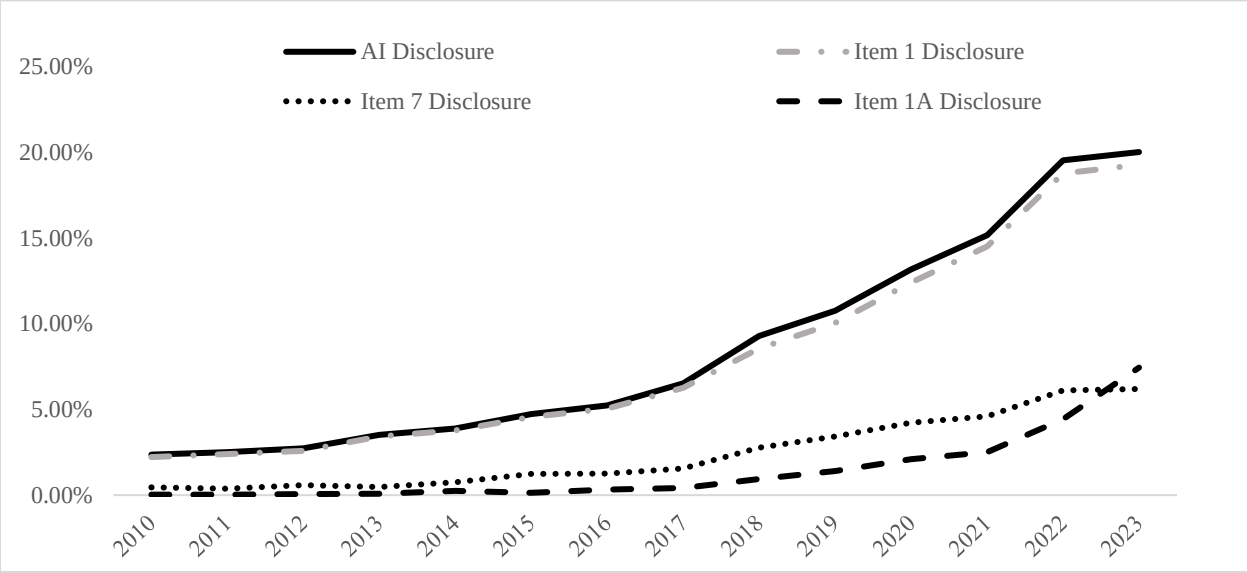
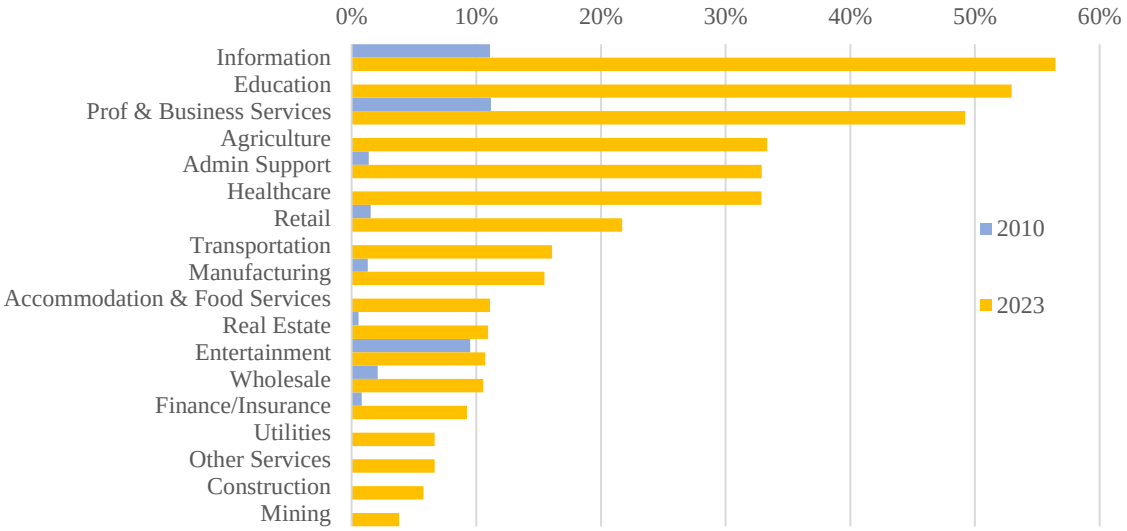


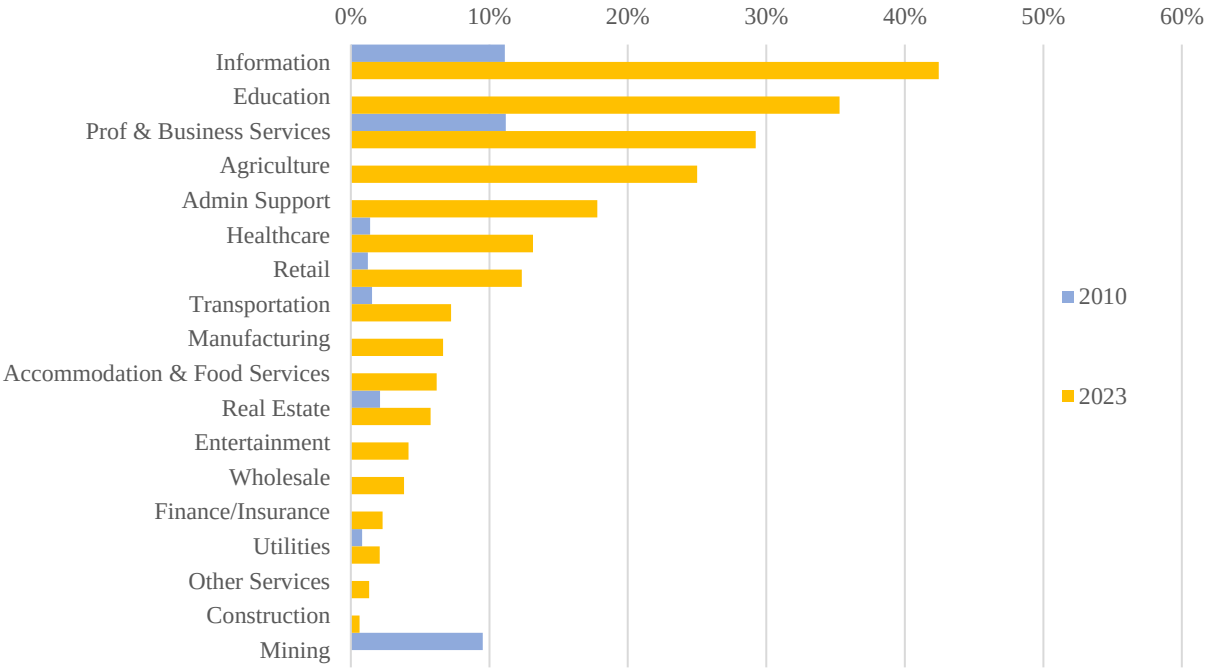
Figure 2: Industrial Distribution of AI Disclosure

This figure shows the industrial distribution of AI disclosures. Panel A, B, C, and D report the distribution based on overall AI disclosure, AI revenue disclosure, AI cost disclosure, and AI risk disclosure, respectively.

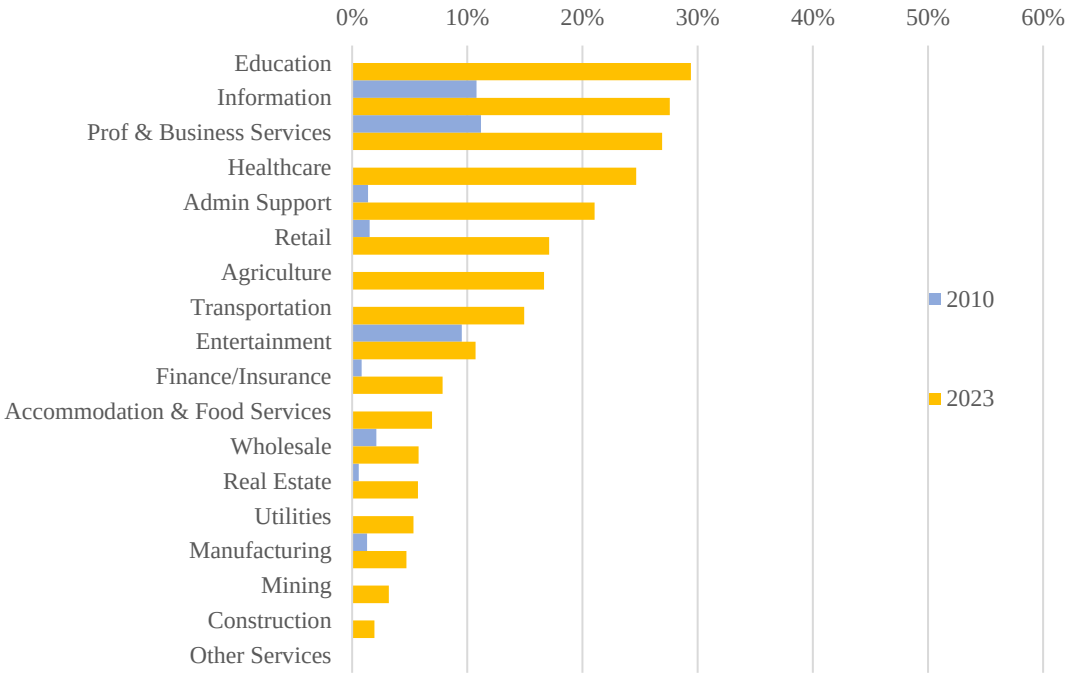
Panel A: AI Disclosure



Panel B: AI Disclosure-Revenue



Panel C: AI Disclosure-Cost



Panel D: AI Risk Disclosure

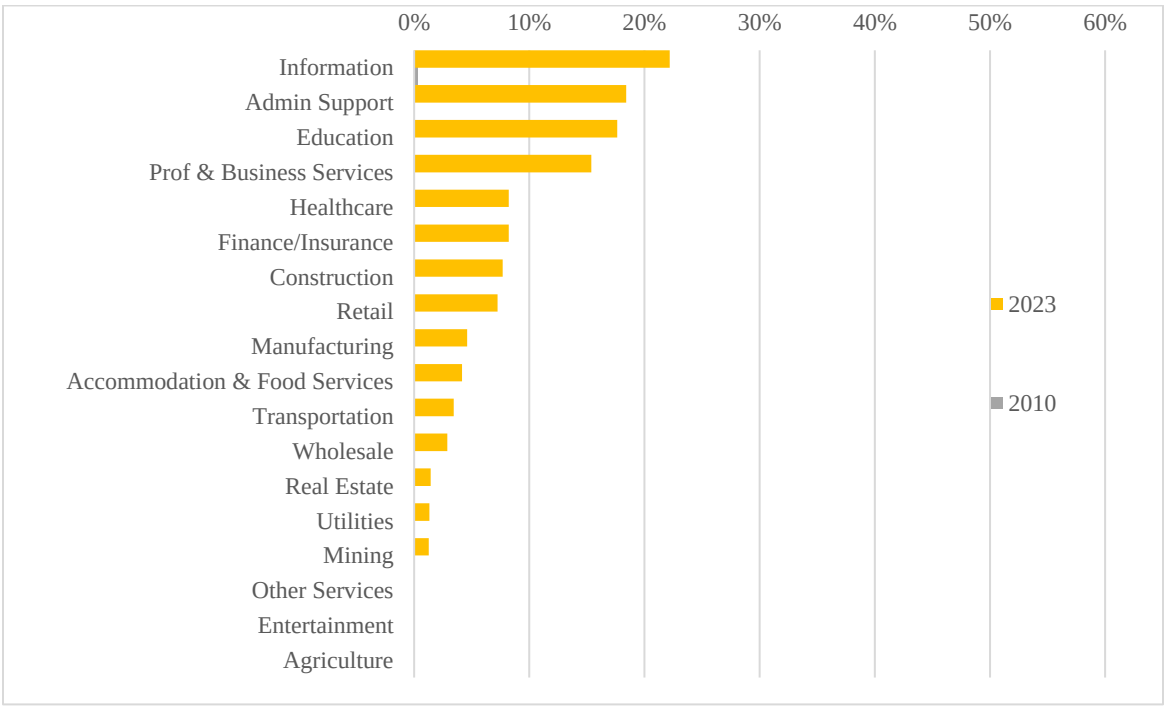


Table 1: Trends of AI Disclosure

Panel A of this table summarizes the number of firms in the sample, the number of firms with AI disclosures (AI Disclosure) in SEC filings, and the categories of AI disclosure for each calendar year. AI revenue (cost) indicates that AI is primarily applied to enhance revenue(cost). AI revenue disclosure includes categories of product development, pricing optimization, and AI product provider. AI cost disclosure includes categories of inventory management and operational efficiency. Panel B of this table summarizes the number of firms in the sample, the number of firms with AI risk disclosures in item 1A (AI Risk Disclosure) in SEC filings, and the categories of AI risk disclosure for each calendar year. The AI risk disclosure categories include regulatory risk, operational risk, competition risk, cybersecurity risk, ethical risk, and third-party risk. The sample covers U.S. public firms from 2010 to 2023.

Panel A: AI disclosure in item 1 and item 7

Year	Total Firms	AI Disclosure	AI Revenue Disclosure	AI Cost Disclosure	Product Development	Pricing Optimization	AI Product Provider	Inventory Management	Operation Efficiency
2010	3,511	83	42	43	17	1	30	3	42
2011	3,460	87	44	43	19	1	29	2	41
2012	3,435	94	51	50	21	1	33	3	47
2013	3,444	121	71	55	26	3	48	2	53
2014	3,550	138	77	68	35	3	44	2	66
2015	3,642	172	102	81	38	4	69	1	80
2016	3,650	191	122	86	50	5	82	1	85
2017	3,606	235	139	111	76	7	74	2	109
2018	3,600	334	212	151	124	4	118	6	148
2019	3,652	393	255	169	155	8	147	6	165
2020	3,670	483	303	229	186	12	182	8	225
2021	3,950	599	380	282	231	21	218	9	275
2022	4,156	812	508	407	335	25	274	12	399
2023	4,166	834	547	411	362	22	284	11	403

Panel B: AI risk disclosure in item 1A

Year	Total Firms	AI Disclosure	AI Risk Disclosure	Regulatory Risk	Operational Risk	Competition Risk	Cybersecurity Risk	Ethical Risk	Third-party Risk
2010	3,511	83	1	0	1	0	0	0	0
2011	3,460	87	1	0	1	0	0	0	0
2012	3,435	94	2	0	2	0	1	0	0
2013	3,444	121	3	1	1	1	1	1	0
2014	3,550	138	9	0	4	3	3	0	0
2015	3,642	172	5	2	2	1	2	0	0
2016	3,650	191	12	4	5	0	1	1	0
2017	3,606	235	15	0	7	4	3	0	0
2018	3,600	334	34	3	14	9	9	1	0
2019	3,652	393	51	4	26	12	12	2	3
2020	3,670	483	77	8	40	14	18	3	3
2021	3,950	599	99	19	49	20	21	13	4
2022	4,156	812	184	51	106	30	29	29	10
2023	4,166	834	310	91	153	46	56	50	22

Table 2: Summary Statistics

This table presents summary statistics of the main variables. The sample period is from 2010 to 2023. All variables are defined in Appendix A. All continuous variables are winsorized at the 1st and 99th percentile.

Panel A: AI-Related Variables

	N	Mean	S.D.	P25	P50	P75
AI Disclosure	47,870	0.094	0.291	0	0	0
AI Disclosure-Revenue	47,870	0.059	0.235	0	0	0
AI Disclosure-Cost	47,870	0.045	0.207	0	0	0
AI Disclosure-Product Development	47,870	0.035	0.183	0	0	0
AI Disclosure-Inventory Management	47,870	0.001	0.037	0	0	0
AI Disclosure-Operational Efficiency	47,870	0.044	0.205	0	0	0
AI Disclosure-Pricing Optimization	47,870	0.002	0.049	0	0	0
AI Disclosure-AI Product Provider	47,870	0.033	0.180	0	0	0
AI Intensity	47,870	0.015	0.049	0	0	0
AI Risk Disclosure	47,870	0.012	0.108	0	0	0
AI Risk Disclosure-Regulation	47,870	0.004	0.063	0	0	0
AI Risk Disclosure-Operation	47,870	0.009	0.094	0	0	0
AI Risk Disclosure-Competition	47,870	0.005	0.071	0	0	0
AI Risk Disclosure-Cybersecurity	47,870	0.004	0.061	0	0	0
AI Risk Disclosure-Ethical	47,870	0.004	0.062	0	0	0
AI Risk Disclosure-Third Party	47,870	0.002	0.044	0	0	0
AI Employee Share	29,130	0.101	0.489	0	0	0

Panel B: Firm Growth and Risk

	N	Mean	S.D.	P25	P50	P75
log(Employees)	47,187	1.184	1.279	0.143	0.700	1.902
log(Sales)	45,623	6.003	2.445	4.478	6.205	7.700
CAPX/Assets	47,870	0.033	0.046	0.003	0.017	0.042
log(R&D)	23,716	2.806	2.147	0.742	2.881	4.331
log(AI Patents)	47,870	0.006	0.083	0	0	0
log(Product Patents)	47,870	0.015	0.144	0	0	0
COGS/Sales	45,655	1.528	6.990	0.360	0.608	0.778
COGS/Employees	45,630	4.243	8.387	0.724	1.644	4.071
Operating Expense/Employees	45,630	6.101	9.573	1.828	3.120	6.341
IVOL	47,870	0.023	0.016	0.013	0.019	0.029
SlopeD	47,870	0.586	0.510	0.209	0.422	0.859
VRP	47,870	0.049	0.248	-0.089	-0.034	0.091
MFIV	47,870	0.447	0.418	0.163	0.301	0.584
MFIS	47,870	-0.437	0.483	-0.755	-0.460	-0.140

Panel C: Control Variables

	N	Mean	S.D.	P25	P50	P75
log(Assets)	47,870	6.745	2.247	5.288	6.892	8.290
ROA	47,870	-0.070	0.304	-0.049	0.015	0.060
Cash	47,870	0.224	0.262	0.036	0.110	0.312
Leverage	47,685	0.245	0.244	0.030	0.179	0.387
Net PPE	46,178	0.202	0.239	0.025	0.102	0.285
MtB	47,779	2.217	2.057	1.051	1.460	2.419
log(Firm Age)	47,464	2.624	1.003	1.946	2.773	3.367
Tobin's Q	47,870	2.862	1.762	1.732	2.392	3.320
Intan/Assets	47,870	0.162	0.204	0.003	0.062	0.272
β_{MKT}	47,870	1.017	0.528	0.664	1.008	1.346
XOPR/Assets	47,870	0.706	0.674	0.206	0.538	0.986

Table 3: Correlation Matrix

This table presents Pearson correlation coefficients across our main independent variables. The sample period is from 2010 to 2023. All variables are defined in Appendix A. All continuous variables are winsorized at the 1st and 99th percentile.

Panel A: AI Disclosure

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) AI Disclosure	1.00									
(2) AI Disclosure Intensity	0.92	1.00								
(3) AI Revenue Disclosure	0.78	0.72	1.00							
(4) AI Cost Disclosure	0.67	0.60	0.14	1.00						
(5) AI Disclosure - Product Development	0.59	0.54	0.76	0.12	1.00					
(6) AI Disclosure - Inventory Management	0.11	0.12	0.02	0.17	0.01	1.00				
(7) AI Disclosure - Operational Efficiency	0.67	0.59	0.14	0.99	0.12	0.04	1.00			
(8) AI Disclosure - Pricing Optimization	0.15	0.13	0.20	0.09	0.04	0.07	0.08	1.00		
(9) AI Disclosure - AI Product Provider	0.58	0.55	0.75	0.15	0.31	0.01	0.16	0.03	1.00	
(10) AI Employee Share	0.24	0.24	0.26	0.09	0.26	0.00	0.09	0.06	0.20	1.00

Panel B: AI Risk Disclosure

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) AI Risk Disclosure	1.00						
(2) AI Risk Disclosure – Regulation	0.58	1.00					
(3) AI Risk Disclosure – Operation	0.87	0.39	1.00				
(4) AI Risk Disclosure – Competition	0.65	0.25	0.68	1.00			
(5) AI Risk Disclosure – Cybersecurity	0.56	0.28	0.54	0.26	1.00		
(6) AI Risk Disclosure – Ethical	0.57	0.43	0.50	0.30	0.23	1.00	
(7) AI Risk Disclosure – Third Party	0.41	0.22	0.41	0.27	0.18	0.28	1.00

Table 4: Determinants of the AI Disclosure.

This table presents results on the determinants of AI disclosure in 10-K filings by estimating the following regression at the firm-year level:

$$AI\ Disclosure_{i,t+1}/AI\ Intensity_{i,t+1} = \beta \cdot AI\ Employee\ Share_{i,t} + \lambda \cdot Other\ Firm\ Chacteristics_{i,t} + D_j + D_t + \epsilon_{i,t+1}$$

In columns (1), (2), and (3), the regression specification is a logistic model, and the dependent variables (*AI Disclosure Vars*) are *AI Disclosure, AI Disclosure-Revenue*, and *AI Disclosure-Cost*. *AI Disclosure* is equal to one if one of the AI keywords is referenced in the 10-K filings and identified as AI-related by ChatGPT. *AI Disclosure-Revenue (Cost)* is equal to one if the AI Disclosure is primarily related to enhancing revenue (cost). In columns (4), (5), and (6), the model specification is a panel regression, and the dependent variables (*AI Disclosure Vars*) are *AI Intensity, AI Intensity-Revenue*, and *AI Intensity-Cost*. *AI Intensity* is measured by the number of AI disclosure words processed by GPT, scaled by the total number of words in the corpus extracted from 10-K filings. *AI Intensity-Revenue (Cost)* is the AI Intensity if the AI Disclosure is primarily related to enhancing revenue (cost). All independent variables are lagged by one period. The sample is from 2010 to 2023. Variable definitions are provided in Table A6. The regression coefficients are reported, followed by the robust t-statistics (in parentheses) based on standard errors clustered on the 5-digit NAICS code. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The industry-fixed effect is based on the 2-digit NAICS code.

Spec.	Disclosure Dummy Logit			Disclosure Intensity Panel		
Dep. Var.	*AI Disclosure*	*AI Disclosure -Revenue*	*AI Disclosure -Cost*	*AI Intensity*	*AI Intensity -Revenue*	*AI Intensity -Cost*
	(1)	(2)	(3)	(4)	(5)	(6)
AI Employee Share	0.588***	0.630***	0.122	0.045***	0.042***	0.009*
	(6.344)	(7.394)	(1.317)	(7.967)	(7.846)	(1.794)
log(Assets)	0.321***	0.249***	0.226***	0.003**	0.002**	0.001*
	(6.207)	(4.554)	(3.965)	(2.535)	(2.121)	(1.888)
ROA	-0.351	-0.555*	0.402	0.004	-0.004	0.008**
	(-1.258)	(-1.893)	(1.037)	(0.849)	(-0.743)	(2.369)
Cash	0.926**	0.625	0.855*	0.006	0.004	0.001
	(2.472)	(1.574)	(1.918)	(0.588)	(0.541)	(0.226)
Leverage	-0.947***	-1.359***	-0.552	-0.026***	-0.019***	-0.010**
	(-2.597)	(-3.156)	(-1.305)	(-4.000)	(-4.448)	(-2.358)
Net PPE	-2.812***	-2.818***	-2.522***	-0.035***	-0.021***	-0.015***
	(-4.951)	(-4.155)	(-3.963)	(-3.590)	(-3.435)	(-2.622)
MtB	0.069**	0.043	0.065	0.003**	0.001	0.001**
	(1.990)	(1.152)	(1.546)	(2.170)	(1.431)	(2.213)
log(Firm Age)	-0.357***	-0.197*	-0.354***	-0.003	-0.001	-0.003**

	(-3.430)	(-1.755)	(-3.091)	(-1.591)	(-0.394)	(-2.234)
N	27,878	25,986	27,358	27,957	27,957	27,957
Adj. R2	0.2066	0.1878	0.1490	0.116	0.1009	0.0411
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster SE	Yes	Yes	Yes	Yes	Yes	Yes

Note: .

Table 5: AI Disclosure and Firm Growth.

This table presents regression results on predicting firm growth using AI disclosure measures at the firm-year level. The dependent variables (*Firm Growt*h) are *log(Employees)*, *log(Sales)*, *CAPX/Assets* and *log(R&D)*. *log(Employees)* is the logarithm of the number of employees. *log(Sales)* is the logarithm of the gross sales. *CAPX/Assets* is Capital expenditures scaled by the total assets. *log(R&D)* is the logarithm of the research and development expenditure. Panel A and B report the results estimating the predictability of AI disclosure variables on the firm growth by estimating the following regression:

$$Firm\ Growth_{i,t+1} = \beta_1 \cdot AI\ Disclosure_{i,t} + \beta_2 \cdot AI\ Employee\ Share_{i,t} + \lambda \cdot Controls_{i,t} + D_j + D_t + \epsilon_{i,t+1}$$

In Panel A*, AI disclosure* is equal to one if the AI keyword is referenced in the 10-K statement and identified as AI-related by ChatGPT. *AI Employee Share* is the fraction of employees classified as AI-related over the total number of employees. In Panel B, *AI Disclosure-Revenue (Cost)* is equal to one if AI Disclosure is primarily related to enhancing revenue (cost). Panel C reports the results of the predictability of AI intensity on the firm growth by estimating the following regression:

$$Firm\ Growth_{i,t+1} = \beta_1 \cdot AI\ Intensity_{\ i,t} + \beta_2 \cdot AI\ Employee\ Share_{i,t} + \lambda \cdot Controls_{i,t} + D_j + D_t + \epsilon_{i,t+1}$$

AI intensity is the number of words in AI disclosure words processed by GPT scaled by the total number of words in the corpus extracted from 10-K filings. *AI Intensity-Revenue (Cost)* is the AI Intensity if the AI Disclosure is primarily related to enhancing revenue (cost). All independent variables are lagged by one period. The sample spans from 2010 to 2023. Variable definitions are provided in Table A6. The regression coefficients are reported, followed by the robust t-statistics (in parentheses) based on standard errors clustered by the 5-digit NAICS code. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The industry-fixed effect is based on the 2-digit NAICS code

Panel A: The Predictability of AI Disclosure on Firm Growth

Dep. Var.	*log(Employees)*		*log(Sales)*		*CAPX/Assets*		*log(R&D)*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AI Disclosure	0.234***	0.139***	0.307***	0.160***	-0.002	0.002***	0.871***	0.464***
	(6.648)	(6.653)	(5.205)	(6.315)	(-1.564)	(3.588)	(12.950)	(11.371)
AI Employee Share	-0.015	-0.042***	-0.003	0.036***	-0.000	0.001**	0.337***	0.156***
	(-0.896)	(-3.850)	(-0.084)	(3.050)	(-1.116)	(2.156)	(8.353)	(8.161)
log(Assets)		0.489***		0.891***		0.000		0.727***
		(76.626)		(142.946)		(0.123)		(56.192)
ROA		-0.565***		1.247***		0.006***		-0.629***
		(-20.951)		(28.563)		(3.674)		(-13.087)
Cash		-0.306***		-1.244***		-0.004***		1.456***
		(-7.163)		(-8.112)		(-2.901)		(22.099)
Leverage		-0.331***		-0.249***		-0.019***		-1.032***

		(-8.891)		(-4.546)		(-11.697)		(-12.419)
Net PPE		-0.007		0.063		0.134***		-1.025***
		(-0.126)		(1.031)		(47.299)		(-9.544)
MtB		0.037***		0.070***		0.002***		0.082***
		(7.347)		(8.166)		(10.344)		(17.168)
log(Firm Age)		0.202***		0.135***		-0.004***		0.002
		(27.520)		(11.972)		(-11.724)		(0.176)
N	28,863	27,747	28,405	27,280	29,130	27,957	15,903	15,206
Adj. R2	0.1665	0.7228	0.1151	0.8687	0.3054	0.5274	0.4077	0.7558
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel B: The Predictability of AI Disclosure-Revenue and AI Disclosure-Cost on Firm Growth

Dep. Var.	*log(Employees)*			*log(Sales)*			*CAPX/Assets*			*log(R&D)*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
AI Disclosure-Revenue	0.021		0.014	0.058**		0.051**	0.002**		0.001*	0.502***		0.497***
	(0.896)		(0.598)	(2.279)		(2.028)	(2.013)		(1.901)	(12.096)		(12.029)
AI Disclosure-Cost		0.257***	0.257***		0.252***	0.250***		0.003***	0.003***		0.267***	0.252***
		(8.808)	(8.804)		(7.730)	(7.772)		(3.381)	(3.325)		(5.207)	(4.986)
AI Employee Share	-0.032***	-0.034***	-0.036***	0.046***	0.047***	0.042***	0.001**	0.001***	0.001**	0.154***	0.192***	0.151***
	(-2.926)	(-3.230)	(-3.270)	(3.773)	(3.747)	(3.497)	(2.389)	(2.683)	(2.282)	(7.909)	(9.382)	(7.893)
log(Assets)	0.490***	0.489***	0.489***	0.892***	0.891***	0.891***	0.000	0.000	0.000	0.729***	0.731***	0.728***
	(76.454)	(76.872)	(76.609)	(142.986)	(143.543)	(143.013)	(0.199)	(0.174)	(0.134)	(56.751)	(57.521)	(56.355)
ROA	-0.563***	-0.567***	-0.567***	1.249***	1.244***	1.245***	0.006***	0.006***	0.006***	-0.624***	-0.633***	-0.626***
	(-20.825)	(-21.126)	(-21.120)	(28.416)	(28.442)	(28.476)	(3.696)	(3.666)	(3.672)	(-12.946)	(-13.555)	(-13.008)
Cash	-0.305***	-0.305***	-0.305***	-1.243***	-1.243***	-1.243***	-0.004***	-0.004***	-0.004***	1.457***	1.459***	1.457***
	(-7.037)	(-7.125)	(-7.130)	(-8.065)	(-8.089)	(-8.098)	(-2.895)	(-2.884)	(-2.890)	(21.939)	(21.796)	(22.032)
Leverage	-0.338***	-0.333***	-0.333***	-0.256***	-0.252***	-0.250***	-0.019***	-0.019***	-0.019***	-1.028***	-1.049***	-1.028***
	(-9.009)	(-8.956)	(-8.926)	(-4.660)	(-4.627)	(-4.573)	(-11.749)	(-11.725)	(-11.707)	(-12.422)	(-12.586)	(-12.404)
Net PPE	-0.018	-0.009	-0.008	0.053	0.059	0.062	0.134***	0.134***	0.134***	-1.035***	-1.054***	-1.027***
	(-0.297)	(-0.154)	(-0.143)	(0.861)	(0.973)	(1.012)	(47.316)	(47.346)	(47.301)	(-9.590)	(-9.675)	(-9.556)
MtB	0.038***	0.037***	0.037***	0.070***	0.070***	0.070***	0.002***	0.002***	0.002***	0.083***	0.083***	0.082***
	(7.488)	(7.413)	(7.386)	(8.198)	(8.140)	(8.148)	(10.391)	(10.403)	(10.354)	(17.298)	(17.340)	(17.168)
log(Firm Age)	0.201***	0.203***	0.203***	0.133***	0.135***	0.135***	-0.004***	-0.004***	-0.004***	-0.000	0.001	0.001
	(27.468)	(27.527)	(27.547)	(11.874)	(12.024)	(12.020)	(-11.786)	(-11.730)	(-11.716)	(-0.013)	(0.078)	(0.087)
N	27,747	27,747	27,747	27,280	27,280	27,280	27,957	27,957	27,957	15,206	15,206	15,206
Adj. R2	0.7222	0.7233	0.7233	0.8684	0.8688	0.8688	0.5273	0.5274	0.5274	0.7554	0.7531	0.7558
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel C: The Predictability of AI Intensity on Firm Growth

Dep. Var.	*log(Employees)*			*log(Sales)*			*CAPX/Assets*			*log(R&D)*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
AI Intensity	0.131***			0.153***			0.002*			0.607***		
	(4.253)			(3.824)			(1.845)			(9.554)		
AI Intensity-Revenue		0.001			0.034			0.001			0.663***	
		(0.023)			(0.816)			(0.662)			(9.421)	
AI Intensity-Cost			0.280***			0.296***			0.003**			0.395***
			(6.098)			(5.603)			(2.301)			(4.956)
AI Employee Share	-0.036***	-0.030***	-0.032***	0.044***	0.049***	0.048***	0.001**	0.001***	0.001***	0.170***	0.168***	0.193***
	(-3.314)	(-2.781)	(-3.060)	(3.613)	(4.000)	(3.885)	(2.504)	(2.629)	(2.738)	(8.679)	(8.499)	(9.374)
log(Assets)	0.489***	0.490***	0.489***	0.892***	0.892***	0.892***	0.000	0.000	0.000	0.730***	0.731***	0.732***
	(76.641)	(76.594)	(76.808)	(143.205)	(143.247)	(143.539)	(0.202)	(0.233)	(0.214)	(57.061)	(57.383)	(57.723)
ROA	-0.564***	-0.563***	-0.566***	1.248***	1.248***	1.246***	0.006***	0.006***	0.006***	-0.630***	-0.626***	-0.633***
	(-20.858)	(-20.834)	(-20.973)	(28.474)	(28.366)	(28.416)	(3.685)	(3.695)	(3.676)	(-13.158)	(-13.016)	(-13.566)
Cash	-0.306***	-0.305***	-0.305***	-1.243***	-1.242***	-1.243***	-0.004***	-0.004***	-0.004***	1.454***	1.457***	1.457***
	(-7.089)	(-7.026)	(-7.083)	(-8.079)	(-8.057)	(-8.075)	(-2.897)	(-2.891)	(-2.890)	(21.987)	(21.866)	(21.783)
Leverage	-0.335***	-0.339***	-0.336***	-0.254***	-0.257***	-0.255***	-0.019***	-0.019***	-0.019***	-1.037***	-1.034***	-1.049***
	(-8.969)	(-9.030)	(-9.002)	(-4.630)	(-4.692)	(-4.670)	(-11.737)	(-11.769)	(-11.744)	(-12.471)	(-12.441)	(-12.604)
Net PPE	-0.014	-0.019	-0.014	0.055	0.050	0.054	0.134***	0.134***	0.134***	-1.039***	-1.044***	-1.058***
	(-0.237)	(-0.313)	(-0.242)	(0.905)	(0.828)	(0.890)	(47.320)	(47.333)	(47.350)	(-9.590)	(-9.617)	(-9.685)
MtB	0.038***	0.038***	0.037***	0.070***	0.071***	0.070***	0.002***	0.002***	0.002***	0.083***	0.083***	0.083***
	(7.445)	(7.517)	(7.458)	(8.198)	(8.206)	(8.159)	(10.392)	(10.422)	(10.417)	(17.198)	(17.316)	(17.329)
log(Firm Age)	0.202***	0.201***	0.202***	0.134***	0.133***	0.134***	-0.004***	-0.004***	-0.004***	0.000	-0.001	0.001
	(27.495)	(27.448)	(27.491)	(11.897)	(11.872)	(11.949)	(-11.776)	(-11.798)	(-11.769)	(0.006)	(-0.101)	(0.041)
N	27,747	27,747	27,747	27,280	27,280	27,280	27,957	27,957	27,957	15,206	15,206	15,206
Adj. R2	0.7224	0.7222	0.7226	0.8685	0.8684	0.8685	0.5273	0.5273	0.5273	0.7546	0.7542	0.7530
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 6: AI Disclosure and Costs

This table presents regression results on predicting the cost of goods sold (COGS) using AI disclosure measures at the firm-year level. The dependent variables are *COGS/Sales, COGS/Employees*, and *Operating Expense/Employees*. *COGS/Sales* is the COGS scaled by total sales. *COGS/Employees* is the COGS scaled by the number of employees. *Operating Expense/Employees* is the operating expenses scaled by the number of employees. Panel A and B report the results estimating the predictability of AI disclosure variables on the COGS by estimating the following regression:

$$COGS_{i,t+1} = \beta_1 \cdot AI\ Disclosure_{i,t} + \beta_2 \cdot AI\ Employee\ Share_{i,t} + \lambda \cdot Controls_{i,t} + D_j + D_t + \epsilon_{i,t+1}$$

In Panel A, *AI disclosure* is equal to one if the AI keyword is referenced in the 10-K statement and identified as AI-related by ChatGPT. *AI Employee Share* is the fraction of employees classified as AI-related over the total number of employees. In Panel B, *AI Disclosure-Revenue (Cost)* is equal to one if AI Disclosure is primarily related to enhancing revenue (cost). Panel C reports the results of the predictability of AI intensity on the COGS by estimating the following regression:

$$COGS_{i,t+1} = \beta_1 \cdot AI\ Intensity_{i,t} + \beta_2 \cdot AI\ Employee\ Share_{i,t} + \lambda \cdot Controls_{i,t} + D_j + D_t + \epsilon_{i,t+1}$$

AI intensity is the number of words in AI disclosure words processed by GPT scaled by the total number of words in the corpus extracted from 10-K filings. *AI Intensity-Revenue (Cost)* is the AI Intensity if the AI Disclosure is primarily related to enhancing revenue (cost). All independent variables are lagged by one period. The sample spans from 2010 to 2023. Variable definitions are provided in Table A6. The regression coefficients are reported, followed by the robust t-statistics (in parentheses) based on standard errors clustered by the 5-digit NAICS code. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The industry-fixed effect is based on the 2-digit NAICS code.

Panel A: The Predictability of AI Disclosure on Corporate Costs

Dep. Var.	*COGS/Sales*		*COGS/Employees*		*Operating Expense/Employees*	
	(1)	(2)	(3)	(4)	(5)	(6)
AI Disclosure	-0.211**	-0.352***	-0.667***	-0.606***	-0.717***	-0.815***
	(-2.481)	(-4.593)	(-5.964)	(-5.725)	(-5.711)	(-7.019)
AI Employee Share	-0.025	-0.389***	-0.112***	-0.081**	0.061	-0.067
	(-0.549)	(-5.367)	(-2.830)	(-2.207)	(1.073)	(-1.195)
log(Assets)		0.138***		0.419***		0.447***
		(4.306)		(9.165)		(9.401)
ROA		-5.809***		-0.807**		-2.543***
		(-8.507)		(-2.362)		(-7.231)
Cash		4.980***		1.708***		3.412***
		(5.464)		(5.133)		(9.349)
Leverage		-0.589***		1.802***		1.611***
		(-3.619)		(4.740)		(4.089)

Net PPE		0.158		0.488		-0.623
		(0.745)		(0.716)		(-0.839)
MtB		-0.182***		-0.196***		-0.169***
		(-2.906)		(-7.913)		(-6.604)
log(Firm Age)		-0.191***		-0.591***		-0.825***
		(-4.316)		(-7.531)		(-10.870)
N	28,408	27,283	28,342	27,255	28,342	27,255
Adj. R2	0.0809	0.2119	0.1512	0.1341	0.1640	0.1618
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster SE	Yes	Yes	Yes	Yes	Yes	Yes

Panel B: The Predictability of AI Disclosure-Revenue and AI Disclosure-Cost on Corporate Costs

Dep. Var.	*COGS/Sales*			*COGS/Employees*			*Operating Expense/Employees*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
AI Disclosure-Revenue	-0.388***		-0.381***	-0.466***		-0.448***	-0.564***		-0.538***
	(-3.929)		(-3.871)	(-4.284)		(-4.149)	(-4.368)		(-4.177)
AI Disclosure-Cost		-0.247***	-0.234***		-0.668***	-0.653***		-0.960***	-0.942***
		(-3.807)	(-3.682)		(-5.539)	(-5.450)		(-7.453)	(-7.345)
AI Employee Share	-0.388***	-0.416***	-0.384***	-0.097**	-0.124***	-0.087**	-0.093*	-0.123**	-0.079
	(-5.344)	(-5.576)	(-5.318)	(-2.566)	(-3.315)	(-2.357)	(-1.667)	(-2.186)	(-1.425)
log(Assets)	0.137***	0.136***	0.138***	0.417***	0.417***	0.419***	0.444***	0.444***	0.447***
	(4.286)	(4.267)	(4.305)	(9.120)	(9.123)	(9.161)	(9.329)	(9.348)	(9.394)
ROA	-5.815***	-5.808***	-5.811***	-0.817**	-0.804**	-0.806**	-2.556***	-2.538***	-2.540***
	(-8.511)	(-8.500)	(-8.508)	(-2.392)	(-2.354)	(-2.359)	(-7.269)	(-7.211)	(-7.219)
Cash	4.978***	4.976***	4.979***	1.707***	1.703***	1.705***	3.410***	3.405***	3.408***
	(5.460)	(5.454)	(5.463)	(5.111)	(5.096)	(5.118)	(9.306)	(9.297)	(9.326)
Leverage	-0.584***	-0.575***	-0.589***	1.816***	1.819***	1.803***	1.632***	1.633***	1.614***
	(-3.605)	(-3.561)	(-3.624)	(4.779)	(4.786)	(4.742)	(4.143)	(4.144)	(4.094)
Net PPE	0.168	0.177	0.159	0.516	0.514	0.492	-0.582	-0.591	-0.617
	(0.789)	(0.832)	(0.752)	(0.758)	(0.755)	(0.722)	(-0.787)	(-0.798)	(-0.832)
MtB	-0.182***	-0.183***	-0.182***	-0.198***	-0.197***	-0.196***	-0.172***	-0.171***	-0.169***
	(-2.914)	(-2.918)	(-2.905)	(-7.977)	(-7.985)	(-7.918)	(-6.695)	(-6.668)	(-6.600)
log(Firm Age)	-0.189***	-0.189***	-0.190***	-0.586***	-0.589***	-0.591***	-0.819***	-0.824***	-0.825***
	(-4.280)	(-4.287)	(-4.304)	(-7.491)	(-7.509)	(-7.528)	(-10.806)	(-10.848)	(-10.871)
N	27,283	27,283	27,283	27,255	27,255	27,255	27,255	27,255	27,255
Adj. R2	0.2119	0.2117	0.2119	0.1338	0.1339	0.1341	0.1613	0.1616	0.1618
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Panel C: The Predictability of AI Intensity on Corporate Costs

Dep. Var.	*COGS/Sales*			*COGS/Employees*			*Operating Expense/Employees*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
AI Intensity	-0.471***			-0.810***			-1.018***		
	(-3.296)			(-5.227)			(-5.679)		
AI Intensity-Revenue		-0.522**			-0.736***			-0.850***	
		(-2.569)			(-4.404)			(-3.928)	
AI Intensity-Cost			-0.416***			-0.845***			-1.154***
			(-3.915)			(-4.549)			(-5.950)
AI Employee Share	-0.399***	-0.398***	-0.417***	-0.099***	-0.104***	-0.128***	-0.094*	-0.104*	-0.130**
	(-5.414)	(-5.377)	(-5.579)	(-2.658)	(-2.754)	(-3.390)	(-1.694)	(-1.868)	(-2.292)
log(Assets)	0.137***	0.136***	0.136***	0.417***	0.416***	0.416***	0.444***	0.442***	0.443***
	(4.279)	(4.266)	(4.261)	(9.126)	(9.103)	(9.102)	(9.339)	(9.306)	(9.311)
ROA	-5.811***	-5.815***	-5.809***	-0.811**	-0.818**	-0.808**	-2.549***	-2.557***	-2.544***
	(-8.507)	(-8.508)	(-8.500)	(-2.375)	(-2.394)	(-2.366)	(-7.250)	(-7.270)	(-7.230)
Cash	4.978***	4.977***	4.976***	1.708***	1.706***	1.704***	3.412***	3.409***	3.407***
	(5.459)	(5.456)	(5.453)	(5.117)	(5.105)	(5.096)	(9.311)	(9.295)	(9.286)
Leverage	-0.582***	-0.579***	-0.574***	1.812***	1.819***	1.825***	1.627***	1.637***	1.642***
	(-3.588)	(-3.580)	(-3.556)	(4.772)	(4.789)	(4.805)	(4.132)	(4.156)	(4.170)
Net PPE	0.170	0.175	0.180	0.510	0.523	0.526	-0.591	-0.573	-0.572
	(0.799)	(0.823)	(0.844)	(0.749)	(0.769)	(0.774)	(-0.798)	(-0.775)	(-0.773)
MtB	-0.182***	-0.183***	-0.183***	-0.197***	-0.198***	-0.198***	-0.171***	-0.172***	-0.172***
	(-2.917)	(-2.925)	(-2.919)	(-7.960)	(-7.998)	(-8.006)	(-6.668)	(-6.721)	(-6.704)
log(Firm Age)	-0.189***	-0.188***	-0.188***	-0.587***	-0.585***	-0.587***	-0.820***	-0.817***	-0.820***
	(-4.281)	(-4.263)	(-4.279)	(-7.501)	(-7.480)	(-7.490)	(-10.822)	(-10.792)	(-10.811)
N	27,283	27,283	27,283	27,255	27,255	27,255	27,255	27,255	27,255
Adj. R2	0.2118	0.2118	0.2117	0.1339	0.1338	0.1338	0.1615	0.1613	0.1613
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 7: AI Risk Disclosure and Firm Risk

This table presents regression results on predicting the firm stock and option-implied risk using AI risk disclosure measures at the firm-year level. The dependent variables are the stock idiosyncratic volatility (*IVOL*), the slopeness of the out-the-money put options' implied volatility on moneyness (*SlopeD*), the variance risk premium defined as the difference between the implied variance from the 30-day options and realized variance of daily stock returns over the same period (*VRP*), the model-free implied volatility for options with a 30-day maturity, computed as the second central moment of the risk-neutral distribution (*MFIV*), and the model-free implied skewness for options with a 30-day maturity, computed as the third central moment of the risk-neutral distribution (*MFIS*). Panel A reports the results estimating the predictability of AI risk disclosure variables on the firm risk variables by estimating the following regression:

$$Firm\ Risk_{i,t+1} = \beta_1 \cdot AI\ Risk\ Disclosure_{i,t} + \lambda \cdot Risk\ Controls_{i,t} + D_j + D_t + \epsilon_{i,t+1}$$

Panel B reports the results estimating the predictability of AI risk disclosure variables on the firm risk variables by estimating the following regression,

$$Firm\ Risk_{i,t+1} = \beta_1 \cdot AI\ Risk\ Disclosure^n_{i,t} + \lambda \cdot Risk\ Controls_{i,t} + D_j + D_t + \epsilon_{i,t+1}$$

where $AI\ Risk\ Disclosure^n_{i,t}$ is the AI risk disclosure for the *n*th category. Each line represents a standalone regression using the categorical AI risk disclosure as the main independent variable. The robust t-statistics (in parentheses) are based on standard errors clustered by the 5-digit NAICS code. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The industry-fixed effect is based on the 2-digit NAICS code.

Panel A: AI Risk Disclosure at the Aggregated Level

Dep. Var.	(1) *IVOL*	(2) *SlopeD*	(3) *VRP*	(4) *MFIV*	(4) *MFIS*
AI Risk Disclosure	0.002***	0.156***	0.098***	0.190***	0.039**
	(3.600)	(7.039)	(10.137)	(12.668)	(2.000)
log(Assets)	-0.003***	-0.011***	-0.049***	-0.101***	-0.106***
	(-67.600)	(-3.388)	(-36.913)	(-46.016)	(-57.001)
log(Firm Age)	-0.002***	-0.017***	-0.011***	-0.051***	-0.027***
	(-15.913)	(-3.330)	(-3.075)	(-8.329)	(-7.694)
Tobin's Q	-0.001***	0.014***	-0.026***	-0.048***	-0.068***
	(-20.220)	(5.192)	(-12.130)	(-13.003)	(-27.569)
Leverage	0.006***	0.062*	0.064***	0.191***	0.138***
	(11.005)	(1.766)	(4.396)	(8.238)	(5.853)
Cash	0.008***	-0.301***	0.173***	0.362***	0.279***
	(12.558)	(-8.819)	(4.750)	(5.882)	(14.457)
Intangible/Assets	-0.001***	0.048*	0.003	-0.056***	-0.178***
	(-3.072)	(1.896)	(0.236)	(-3.005)	(-11.002)
Market Beta	0.004***	-0.104***	-0.058***	0.043***	0.029***
	(21.075)	(-8.121)	(-8.364)	(3.708)	(4.178)
Operating Expense	0.001***	-0.065***	-0.051***	-0.060***	-0.029***
	(7.596)	(-7.056)	(-10.402)	(-7.660)	(-5.849)
Intercept	0.041***	0.869***	0.564***	1.365***	0.643***
	(61.444)	(20.795)	(24.233)	(37.955)	(24.639)
N	47,870	47,870	47,870	47,870	47,870
Adj. R2	0.3522	0.0685	0.2176	0.3393	0.2419
Industry FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes	Yes	Yes

Panel B: Categorical AI Risk Disclosures

Dep. Var.	(1) *IVOL*	(2) *SlopeD*	(3) *VRP*	(4) *MFIV*	(4) *MFIS*
AI Risk Disclosure - Regulation	0.003***	0.171***	0.140***	0.243***	0.070**
	(3.112)	(4.708)	(8.898)	(8.932)	(2.343)
AI Risk Disclosure - Operation	0.002***	0.147***	0.098***	0.200***	0.042*
	(3.932)	(5.433)	(8.876)	(11.700)	(1.843)
AI Risk Disclosure - Competition	0.001	0.191***	0.095***	0.191***	-0.033
	(1.518)	(5.035)	(7.295)	(9.189)	(-1.155)
AI Risk Disclosure - Cybersecurity	0.002**	0.115***	0.069***	0.162***	0.046
	(2.544)	(3.386)	(4.711)	(7.408)	(1.544)
AI Risk Disclosure - Ethical	0.003***	0.174***	0.118***	0.240***	0.068*
	(2.895)	(4.374)	(7.221)	(7.419)	(1.946)
AI Risk Disclosure - Third Party	0.003**	0.052	0.099***	0.216***	0.062*
	(2.054)	(1.106)	(3.495)	(5.524)	(1.750)
N	47,870	47,870	47,870	47,870	47,870
Controls	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes	Yes	Yes

Appendix

Table A1: Examples of AI-related disclosure in 10-K statements.

International Business Machines (IBM), 2022, 10-K Item 1 Business

This was signaled almost 100 years ago, in 1924, when C-T-R changed its name to International Business Machines Corporation. And it continues today we create sustained value for clients by helping them leverage the power of hybrid cloud and **artificial intelligence** (AI). Our hybrid cloud platform and AI technology support clients' digital transformations and helps them reimagine critical workflows, at scale, and modernize applications to increase agility, drive innovation and create operational efficiencies. Our offerings draw from leading IBM capabilities in software, consulting services capability to deliver business outcomes, and deep incumbency in mission-critical infrastructure, all bolstered by one of the world's leading research organizations.

Blackberry, Ltd., 2022, 10-K Item 1A Risk Factors

The process of developing new technology is complex and uncertain, and involves time, substantial costs and risks, which are further magnified when the development process involves multiple operating platforms. The development of next-generation technologies that utilize new and advanced features, including **artificial intelligence** and machine learning, involves making predictions regarding the willingness of the market to adopt such technologies over legacy solutions. The Company may be required to commit significant resources to developing new products, software and services before knowing whether such investment will result in products or services that the market will accept.

Codexis, Inc., 2018, 10-K Item 7 Management Discussion and Analysis

Over the last fifteen years, we have made substantial investments in the development of our CodeEvolver protein engineering technology platform, the primary source of our competitive advantage. Our technology platform is powered by proprietary, **artificial intelligence**-based, computational algorithms that rapidly mine our large and continuously growing library of protein variants performance attributes. These computational outputs enable increasingly reliable predictions for next generation protein variants to be engineered, enabling delivery of targeted performance enhancements in a time-efficient manner.

Table A2: Keywords of AI Disclosure in the 10-K Filings

Keyword	(%) All 10-K Filings	(%) AI Disclosing Transcripts
artificial intelligence	5.91%	40.42%
machine learning	5.13%	35.09%
business intelligence	4.29%	29.35%
big data	2.40%	16.45%
data science	1.84%	12.57%
AI/ML	1.48%	10.10%
data mining	0.98%	6.70%
data scientist	0.93%	6.34%
deep learning	0.49%	3.35%
computer vision	0.45%	3.08%
natural language processing	0.40%	2.75%
chatbot	0.34%	2.33%
image recognition	0.18%	1.21%
object recognition	0.06%	0.39%
large language model	0.04%	0.31%
machine translation	0.04%	0.25%
support vector machine	0.03%	0.23%
classification algorithm	0.03%	0.23%
generative ai	0.03%	0.19%
A.I.	0.02%	0.16%
neutral network	0.02%	0.12%
supervised learning	0.01%	0.07%
computational linguistic	0.01%	0.07%
clustering algorithms	0.01%	0.05%
recommender system	0.01%	0.04%
dimensionality reduction	0.00%	0.03%
information extraction	0.00%	0.03%
kernel method	0.00%	0.01%
unsupervised learning	0.00%	0.01%

Note: The table lists all keywords in the textual analysis for identifying the AI disclosure in the corporate annual report in the first column. The second column summarizes the fraction of occurrence of the corresponding keyword in all 10-K filings in our sample. The second column summarizes the fraction of occurrence of the corresponding keyword in transcripts with AI disclosure. The sample contains U.S. public firms spanning from 2010 to 2023.

Table A3: Most frequently mentioned AI keywords.

Year	Top #1 Frequent	Top #2 Frequent	Top #3 Frequent
2010	business intelligence	AI/ML	data mining
2011	business intelligence	AI/ML	data mining
2012	business intelligence	AI/ML	data mining
2013	business intelligence	AI/ML	data mining
2014	business intelligence	big data	AI/ML
2015	business intelligence	big data	AI/ML
2016	business intelligence	big data	AI/ML
2017	business intelligence	big data	machine learning
2018	artificial intelligence	machine learning	business intelligence
2019	artificial intelligence	machine learning	business intelligence
2020	artificial intelligence	machine learning	business intelligence
2021	artificial intelligence	machine learning	business intelligence
2022	artificial intelligence	machine learning	data science
2023	artificial intelligence	machine learning	data science

Note: The table lists the top 3 keywords with the highest frequency being mentioned in the corporate 10-K report each year. The sample contains U.S. public firms spanning from 2010 to 2023.

Table A4: GPT prompt for the AI Disclosure classification and categorization

Objective:

As an expert financial analyst, review the company's annual 10-K financial statement provided below. Your task is to assess the relevance of the disclosure to artificial intelligence (AI). Determine whether the disclosure is "Related to AI" or "Not Related to AI" and provide a probability score between 0 and 1, indicating the likelihood of the disclosure being related to AI.

Instructions:

Classification:

Label the disclosure as either "Related to AI" or "Not Related to AI".

Probability Score:

Assign a score from 0 to 1 estimating the likelihood of the disclosure being related to AI. A score of 0 means no relation, while 1 indicates a high certainty of relation to AI. The probability score should reflect how clearly the disclosure outlines the company's AI usage.

General Assessment:

If classified as "Not Related to AI", provide reasons supporting this classification.

If classified as "Related to AI", proceed with the following steps:

Application Categorization: Categorize the AI usage into one or more of the following categories based on the description:

- Product Development
- Inventory Management
- Operational Efficiency
- Pricing Optimization
- AI Product Provider

Detailed Reasons: For each category selected, explain how the disclosure aligns with these specific AI applications. Don't infer the application, just show the categories that are explicitly mentioned.

Probability Scores for Subcategories: Provide a score from 0 to 1 for each selected subcategory indicating the likelihood that the disclosure falls into that category.

Definitions:

Related to AI:

This encompasses any mention or implication of AI technologies, such as machine learning, neural networks, deep learning, natural language processing, or AI-driven analytics, and their application in the company's operations, products, or services.

Example: "The company integrates machine learning algorithms to enhance customer personalization on its platforms."

- Product Development

The use of AI in the research, creation, design, or enhancement of the company's products or services. This includes employing AI for R&D, product testing, or accelerating the innovation process.

Example: "AI is utilized in the R&D phase to simulate product performance under various conditions, improving design and reducing time-to-market."

- Inventory Management

AI applications focused on optimizing inventory processes, including managing stock levels, predicting future inventory needs, or streamlining supply chain operations.

Example: “Implementing AI-driven demand forecasting to automate inventory replenishment and reduce overstock.”

- Operational Efficiency

Leveraging AI to improve internal processes and overall productivity. This includes automating routine tasks, predictive maintenance, optimizing workflows, and enhancing day-to-day operational activities.

Example: “Using AI to automate customer service interactions, reducing response times and operational costs.”

- Pricing Optimization

Utilizing AI to develop and implement strategic pricing models. This involves dynamic pricing, competitive pricing analysis, and AI-driven algorithms to adjust prices in real-time to maximize revenue or market share.

Example: “AI algorithms adjust product pricing dynamically based on real-time market data to maintain competitive advantage.”

- AI Product Provider

The company's role as a provider of AI-based products or services. This includes developing, selling, or licensing AI technologies, platforms, or solutions to other businesses or consumers.

Example: “The company offers an AI-powered analytics platform that enables clients to gain actionable insights from their data.”

Output in JSON Response Format:

If Not Related to AI:

```
{
  "classification": "Not Related to AI",
  "probability_score": [score],
  "reasons": "[explanation]"
}
```

If Related to AI:

```
{
  "classification": "Related to AI",
  "probability_score": [score],
  "summary": "[general application summary]",
  "applications": [
    {
      "category": "[Category Name]",
      "probability_score": [score],
      "reasons_of_application": "[explanation for this categorization]"
    }
    // Additional categories as applicable
  ]
}
```

Table A5: ChatGPT prompt for AI intensity

System prompt:

As an expert financial analyst, review the texts from company's annual 10-K financial statement provided below. Your task is to remove contents that are not related to artificial intelligence (AI). When editing, assume that sentences between two segments related to AI also part of this category and should not be removed. Edits should only affect the beginning or end of the provided text. Output back text related to AI. Do not change the text.

User prompt (Example of feed):

Strategies have helped CareAdvantage clients to more effectively: Identify and quantify disease burden and associated risk with their entire population and sub-populations; Improve member care quality through the defensible evaluation of health care providers and facilities; Facilitate provider cooperation and collaboration based on case mix and severity-adjusted data; Forecast resource consumption based on disease burden; and Optimize allocation of resources. Recently, the Company has also assisted health plan clients with respect to further validating the value that they bring to purchasers and have had a demonstrable impact on sales retention and attraction of new sales. The RPNavigator suite of services that the Company offers our state government customers provides opportunities in transparency, improving quality, controlling costs and providing information to consumers and healthcare policymakers to make informed decisions. In addition, this expertise has been applied to help employers assess the efficiency and effectiveness of their present health care insurers, carriers and supporting vendors. CareAdvantage achieves this by empowering employers with the information, skills and guidance necessary to facilitate future purchasing decisions and optimize managerial and administrative practices. In order to deliver these solutions, CareAdvantage utilizes experienced health plan executives and medical directors as well as a wide range of care management operations, clinical data analysis and information technology (IT) subject-matter experts. It is this vast array of experience that enables CareAdvantage to benefit clients with objective and quantifiable insight to develop the strategies and tactical initiatives that combine care management processes with a deep understanding of medical and health care insurance-related best practices. Operations The Company utilizes a multi-disciplinary team approach in providing its management, data analysis and consulting services. The Company, through its employees and independent contractors, assesses care management operations, systems resources, integration and outcomes. Typically, assessment occurs on the client s site, through interviews and data analysis. At the center of CareAdvantage s data-driven analyses is RPNavigator, the next generation software solution. RPNavigator categorizes and quantifies a population s disease burden and provides a clear picture of the health status and severity associated with its clients member populations. RPNavigator s underlying infrastructure incorporates classification methodologies from 3M Health Information Systems along with various analytical techniques to stratify the population and describe the individual member s associated risks in intuitive ways. It also enables the valid assessment of existing health care quality and cost as well as projection of future risk from a resource

consumption, disease progression and mortality perspective. RPNavigator includes a data mining tool (RPN 3 that employs multi-dimensional cubes (data structures) for online analytic processing (OLAP). RPN 3 references the same data set within RPNavigator and allows power users additional flexibility in querying that data. RPNavigator utilizes this information to stratify its clients members, groups and providers through the use of a wide range of clinical and demographic descriptors to quantify their risk as well as evaluate the impact of key interventions and programs. These descriptors and the underlying logic increase the associated transparency of the resulting analyses and support the new direction of the industry. Among the benefits of this solution is the ability of CareAdvantage clients to: Access meaningful information via an Internet-based portal; Track population and member-related changes in disease status and severity over time; Compare client sub-populations; Profile provider using case mix and severity-adjusted techniques; Select and prioritize members who would best benefit from care management interventions; 4 Understand adverse selection associated with existing and/or newly-obtained business as well as understand the impact of a plan s overall turnover in terms of stayers and leavers; and Reduce the dependence on internal resources to develop and produce required reports to accomplish these tasks. There are several related efforts that result in additional sources of income for CareAdvantage, including: Developing new standardized analyses on a client-specific basis to meet a particular need for that client; Undertaking broader analytic consulting projects, using the methodology and logic within RPNavigator, on behalf of clients that need CareAdvantage s expertise in analyzing and interpreting the data; Offering experienced health care executives for care management program leadership, internal physician review services, and mentoring of less experienced health plan staff Executive and Clinical Management Services). CareAdvantage also supports its clients through the provision of a wide range of consulting services to develop and implement the right solutions. These solutions provide the health care industry with strategies and tactical initiatives for effectively managing health care consumption, reducing costs and improving the quality and cost benefit of care. For its services, the Company seeks to be compensated either (i) on a fee-for-service basis; (ii) per member per month (PMPM); or (iii) on the basis of a combination of both fee-for-service and PMPM.

Assistant prompt (Expected response):

The Company, through its employees and independent contractors, assesses care management operations, systems resources, integration and outcomes. Typically, assessment occurs on the client s site, through interviews and data analysis. At the center of CareAdvantage s data-driven analyses is RPNavigator, the next generation software solution. RPNavigator categorizes and quantifies a population s disease burden and provides a clear picture of the health status and severity associated with its clients member populations. RPNavigator s underlying infrastructure incorporates classification methodologies from 3M Health Information Systems along with various analytical techniques to stratify the population and describe the individual member s associated risks in intuitive ways. It also enables the valid assessment of existing health care quality and cost as well as projection of future risk from a resource consumption, disease progression and mortality perspective. RPNavigator includes a data mining tool (RPN 3 that employs multi-dimensional

cubes (data structures) for online analytic processing (OLAP). RPN 3 references the same data set within RPNavigator and allows power users additional flexibility in querying that data. RPNavigator utilizes this information to stratify its clients members, groups and providers through the use of a wide range of clinical and demographic descriptors to quantify their risk as well as evaluate the impact of key interventions and programs. These descriptors and the underlying logic increase the associated transparency of the resulting analyses and support the new direction of the industry. Among the benefits of this solution is the ability of CareAdvantage clients to: Access meaningful information via an Internet-based portal; Track population and member-related changes in disease status and severity over time; Compare client sub-populations; Profile provider using case mix and severity-adjusted techniques; Select and prioritize members who would best benefit from care management interventions; Understand adverse selection associated with existing and/or newly-obtained business as well as understand the impact of a plan s overall turnover in terms of stayers and leavers; and Reduce the dependence on internal resources to develop and produce required reports to accomplish these tasks.

Table A6: GPT prompt for the AI risk classification and categorization

Objective:

As an expert financial analyst, review the risk-related disclosure from company's annual 10-K financial statement provided below. Your task is to assess the relevance of the disclosure to artificial intelligence (AI). Determine whether the disclosure is "Related to AI" or "Not Related to AI" and provide a probability score between 0 and 1 indicating the likelihood of the disclosure being related to AI.

Instructions:

Classification:

Label the disclosure as either "Related to AI" or "Not Related to AI".

Probability Score:

Assign a score from 0 to 1 estimating the likelihood of the disclosure being related to AI. A score of 0 means no relation, while 1 indicates a high certainty of relation to AI. The probability score should reflect how clearly the disclosure outlines the company's AI risks.

General Assessment:

If classified as "Not Related to AI", provide reasons supporting this classification.

If classified as "Related to AI", proceed with the following steps:

Risk Categorization: Categorize the AI disclosure into one or more of the following categories based on the description:

- Regulatory risks
- Operational risks
- Competitive risks
- Cybersecurity risks
- Ethical risks
- Third-party risks

Detailed Reasons: For each category selected, explain how the disclosure aligns with these specific AI Risks. Don't infer the risks just show the categories that are explicitly mentioned.

Probability Scores for Subcategories: Provide a score from 0 to 1 for each selected subcategory indicating the likelihood that the disclosure falls into that category.

Definitions:

Related to AI:

This encompasses any risks of AI technologies, such as machine learning, neural networks, deep learning, natural language processing, data mining or AI-driven analytics, in the company's operations, products, or services.

- Regulatory risks

Regulatory risks related to the regulation of AI and other emerging technologies were broadly cited across many industries. These risk factors expressed concerns about increased compliance and legal risks associated with evolving AI laws, regulations, and standards.

Example: "As with many technological innovations, artificial intelligence presents risks and challenges that could affect its adoption, and therefore our business. Uncertainty in the legal regulatory regime relating to AI may require significant resources to modify and maintain business practices to comply with U.S. and non-U.S. laws, the nature of which cannot be determined at this time. Several jurisdictions around the globe, including Europe and certain U.S. states, have already proposed or enacted laws governing AI. For example, on October 30, 2023, the Biden administration issued an Executive Order to, among other things, establish extensive new standards for AI safety and security. Other jurisdictions may decide to adopt similar or more restrictive legislation that may render the use of such technologies challenging. These obligations may make it harder for us to conduct our business using AI, lead to regulatory fines or penalties, require us to change our product offerings or business practices, or prevent or limit our use of AI. If we cannot use AI, or that use is restricted, our business may be less efficient, or we may be at a competitive disadvantage. Any of these factors could adversely affect our business, financial condition, and results of operations."

- Operational risks

Operational risk factors emphasized the impact of unpredictable disruptions, technical challenges and errors in AI projects that could affect the companies' financial results and business operations.

Example: "The development, adoption, and use for generative AI technologies are still in their early stages and ineffective or inadequate AI development or deployment practices by [the Company] or third-party developers or vendors could result in unintended consequences. For example, AI algorithms that we use may be flawed or may be based on datasets that are biased or insufficient. In addition, any latency, disruption, or failure in our AI systems or infrastructure could result in delays or errors in our offerings. Developing, testing, and deploying resource-intensive AI systems may require additional investment and increase our costs."

- Competitive risks:

Competitive risks refer to the potential loss of market position and adverse business impacts resulting from delays in adopting, investing in, or integrating AI technologies, particularly in fast-evolving sectors like financial services and technology.

Example: "Our industry is marked by rapid technological developments and innovations (such as the use of artificial intelligence and machine learning) and evolving industry standards. If we are unable to provide enhancements and new features and integrations for our existing platform, develop new products that achieve market acceptance, or innovate quickly enough to keep pace with these rapid technological developments, our business could be harmed."

- Cybersecurity Risks:

Cybersecurity Risks refer to potential threats and evolving cyberattack techniques that arise from the use of AI and other emerging technologies.

Example: "As the Company pursues its strategy to grow through acquisitions and to pursue new initiatives that improve our operations and cost structure, the Company is also expanding and improving its information technologies, resulting in a larger technological presence, utilization of "cloud" computing services, and corresponding exposure to cybersecurity risk. Certain new technologies, such as use of autonomous vehicles, remote-controlled equipment, virtual reality, automation and artificial intelligence, present new and significant cybersecurity safety risks that must be analyzed and addressed before implementation. If we fail to assess and identify cybersecurity risks associated with acquisitions and new initiatives, we may become increasingly vulnerable to such risks."

- Ethical risks:

Ethical risks refer to potential reputational harm, liability, and increased costs arising from social and ethical concerns, such as impacts on human rights, privacy, and employment, associated with AI use.

Example: "Social and ethical issues relating to the use of new and evolving technologies such as artificial intelligence ("AI") in our offerings, may result in reputational harm and liability, and may cause us to incur additional research and development costs to resolve such issues. We are increasingly building AI into many of our offerings. As with many innovations, AI presents risks and challenges that could affect its adoption, and therefore our business. If we enable or offer solutions that draw controversy due to their perceived or actual impact on society, we may experience brand or reputational harm, competitive harm or legal liability. Potential government regulation related to AI use and ethics may also increase the burden and cost of research and development in this area, and failure to properly remediate AI usage or ethics issues may cause public confidence in AI to be undermined, which could slow adoption of AI in our products and services. The rapid evolution of AI will require the application of resources to develop, test and maintain our products and services to help ensure that AI is implemented ethically in order to minimize unintended, harmful impact."

- Third-party risks:

Third-party risks refer to the potential challenges and negative impacts, such as increased costs, reputational damage, security vulnerabilities, and regulatory scrutiny, arising from reliance on third-party AI systems and service providers.

Example: "We have established a self-service development environment in which such third-party developers integrate their AI models onto our platform, and we will be dependent in part upon their ability to do so effectively and quickly. We may not have full control over the quality and performance of third-party providers, and therefore, any unexpected deficiencies or problems arising from these third-party providers may cause significant interruptions in the operation of our platform. The failure of third-party developers to integrate their AI models

seamlessly into our platform and/or provide reliable, scalable services may impact the reliability of our platform and harm our reputation and business, results of operations and financial condition."

Output in JSON Response Format:

If Not Related to AI:

```
{
  "classification": "Not Related to AI",
  "probability_score": [score],
  "reasons": "[explanation]"
}
```

If Related to AI:

```
{
  "classification": "Related to AI",
  "probability_score": [score],
  "summary": "[general risk summary]",
  "risks": [
    {
      "category": "[Category Name]",
      "probability_score": [score],
      "reasons_of_risk": "[explanation for this categorization]"
    }
    // Additional categories as applicable
  ]
}
```

Table A7: Variable Definitions.

Variable	Definition
AI Disclosure	An indicator with value of one when there is AI-related disclosure in the item 1 and item 7 sections from the 10-K statement and zero otherwise. The AI disclosure is measured by at least one keyword and processed by the ChatGPT.
AI Disclosure-Revenue	An indicator with value of one when AI disclosure in the 10-K statement is related to revenue enhancement.
AI Disclosure-Cost	An indicator with value of one when AI-related disclosure in the 10-K statement is related to cost enhancement.
AI Intensity	The number of words in AI disclosure words processed by GPT scaled by the total number of words in the corpus extracted from 10-K filings.
AI Intensity-Revenue	AI intensity in the AI Revenue disclosure sample.
AI Intensity-Cost	AI intensity in the AI Cost disclosure sample.
AI Risk Disclosure	An indicator with value of one when there is AI-related risk factor disclosure in the item 1A section from the 10-K statement and zero otherwise. The AI risk disclosure is measured by at least one keyword and processed by the ChatGPT.
AI Employee Share	Following Babina et al. (2024), the fraction of employees classified as AI-related over the total number of employees, scaled by 100.
log(Employees)	The logarithm of the number of employees (EMP on Compustat) in thousands in the fiscal year.
log(Sales)	The logarithm of the gross sales level (SALE on Compustat) in millions in the fiscal year.
CAPX/Assets	Capital expenditures (CAPX on Compustat) scaled by the beginning period of total assets (AT on Compustat) in the fiscal year.
log(R&D)	The logarithm of the research and development expenditure (XRD on Compustat) in millions in the fiscal year.
COGS/Sales	The COGS (COGS on Compustat) scaled by total sales (SALE on Compustat).
COGS/Employees	The COGS (COGS on Compustat) scaled by the number of employees (EMP on Compustat).
Operating Expense/Employees	The operating expenses (XOPR on Compustat) scaled by the number of employees (EMP on Compustat).
IVOL	
SlopeD	The slopeness of the function that relates implied volatility to the Black-Scholes delta for OTM put options (with deltas between -0.5 and -0.1) with a 30-day maturity. The annual measure is constructed by averaging daily values.
VRP	Variance risk premium is defined as the daily difference between the implied variance of returns from t to t+30 calendar days and realized variance of daily returns over the same period (t, t+30). The annual measure is constructed by averaging daily values.
MFIV	Model-free implied volatility for options with a 30-day maturity, calculated as the second central moment of the risk-neutral distribution.

	We follow Bakshi, Kapadia, and Madan (2003) to construct the variable. The variable is constructed at the annual level by taking the average of daily values.
MFIS	Model-free implied skewness for options with a 30-day maturity, calculated as the third central moment of the risk-neutral distribution, normalized by the risk-neutral variance (raised to the power of 3/2). We follow Bakshi, Kapadia, and Madan (2003) to construct the variable. The variable is constructed at the annual level by taking the average of daily values.
log(Assets)	The logarithm of total assets (AT on Compustat) in millions.
ROA	The ratio of net income (NI on Compustat) over the total assets (AT on Compustat) in the fiscal year.
Cash	Cash holdings (CHE on Compustat), scaled by total assets (AT on Compustat).
Leverage	The book value of debt (DLC + DLTT on Compustat) divided by total assets (AT on Compustat).
Net PPE	Net Property, plant, and equipment (PPENT on Compustat), scaled by total assets (AT on Compustat).
MtB	The ratio of the equity market value over the difference between common equity and preferred stock capital at fiscal year-end.
log(Firm Age)	The logarithm of the years between the firm's first year of data in Compustat and the current fiscal year.
Tobin's Q	(Total assets (AT on Compustat) - total common equity (CEQ on Compustat) + share price (PRCC on Compustat) × common shares outstanding (CSHO on Compustat)) / total assets (AT on Compustat).
Intan/Assets	The intangible assets (INTAN on Compustat) scaled by the beginning of the period total assets (AT on Compustat).
β_{MKT}	Sensitivity of daily stock returns to daily S&P returns. We run daily regressions of excess returns on a constant and the market factor for each firm and year. For each firm-by-year combination, β_{MKT} (market beta) corresponds to the estimated regression coefficient.
XOPR/Assets	The operating expense (XOPR on Compustat) scaled by the beginning of the period total assets (AT on Compustat).